Compliance Costs of Contract Regulation*

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Abstract

We estimate the compliance costs of insurance contract regulation in the U.S. property-liability insurance industry by exploiting the rich cross-sectional and time-series variation in the regulation of policy forms. We find that prior approval policy form regulation is associated with higher general expenses. Analyzing changes in regulation, we find that deregulation led to a 1.2 percentage point reduction in expenses for commercial lines and a 1.8 percentage point reduction for personal lines. We then examine broader market impacts. We find significant economies of scale: the largest operations experience negligible compliance burdens compared to small operations. The costs also differ across lines; personal lines costs are more than four times those of conventional commercial lines. Prior approval regulation also delays an insurer's ability to use policy forms. Finally, we find that electronic filing innovation reduces administrative filing costs but does not mitigate the legal compliance costs inherent in prior approval regulation.

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1 Introduction

Insurance contracts are inherently complex. The complexity of insurance policies makes them difficult for many policyholders to understand, creating an informational asymmetry between consumers and insurers. As a result, insurance contracts are often regulated to address the imbalance between consumers and financial institutions (Tennyson, 2010). The informational imbalance between consumers and insurers, however, is not equal across all policyholders. Commercial policyholders, for example, are generally more sophisticated about insurance and contracts than personal lines policyholders. In addition, commercial policyholders use intermediaries (agents or brokers) more frequently than personal lines policyholders.[1] Despite these differences, both personal and commercial insurance contracts are subject to policy form regulation. From 1992 to 2014, 75% of personal lines insurance contracts and 52% of commercial lines insurance contracts were subject to prior approval by regulators. Complying with prior approval regulation of insurance policy forms is costly for insurers, requiring payment of filing fees and hiring staff, lawyers, and consultants to ensure regulatory compliance. The financial costs associated with regulating insurance contracts are subject to tremendous policy debate (Harrington, 2009), yet these costs have been understudied.

While economic theories of regulation – ranging from those focused on correcting market failures (e.g., Public Interest Theory) to those highlighting industry influence (e.g., Regulatory Capture) – agree that regulation imposes costs, there are currently no estimates of the magnitude of the costs of complying with insurance contract regulations. The quantification of compliance costs is a necessary first step in understanding insurance contract regulation, as it provides the essential cost input for cost-benefit analysis and lays the empirical groundwork for subsequent studies to examine how compliance burdens may affect firm behavior, market competition, and consumer prices.[2] To fill this gap, we provide the first empirical estimates of compliance costs imposed by insurance contract regulation.

[1]In 2022, 66.2 percent of the net premiums written in personal lines insurance were direct from the insurer. In the same year, 75.4 percent of the net premiums written in commercial lines were distributed using intermediaries. (Source:Insurance Information Institute (2024)).

[2]If the cost of complying with contract regulation is de minimis, then investigating how compliance affects firm behavior, market competition, and consumer prices is unnecessary. If it does not cost insurers much, then it is not going to affect insurance markets. However, if the costs are consequential, then compliance may affect firm behavior, market competition, and consumer prices. Thus, measuring the cost of compliance is a necessary first step.

The U.S. property and liability insurance industry provides an ideal laboratory to study the costs of contract regulation. (In the insurance industry, the regulation of contracts is also referred to as “policy form regulation” or “form regulation”). In contrast to other financial sectors that are subject to federal regulation, the insurance industry is regulated at the state level. This study exploits the rich cross-sectional and time-series heterogeneity in state regulation to identify the costs of complying with form regulation. There are four sources of variation in the regulation of forms. First, many insurers operate in multiple states, providing insurance in jurisdictions with and without prior approval form regulation in the same line of business and the same year. Second, many insurers operate in multiple lines (e.g., private passenger auto liability, workers compensation, medical professional liability), providing insurance in lines with and without prior approval form regulation within the same state and year. Third, unlicensed insurers (surplus lines insurers) are exempt from form regulation. Fourth, states modify the statutory intensity of their form regulation over time.

We begin our analysis by quantifying the additional costs of complying with prior approval form regulation at the insurer-line level. We find statistically significant differences in costs. Compared to an insurer-line that only writes in states that do not require prior approval, an insurer-line that only writes in states that require prior approval is associated with a one percentage point higher general expenses. This increase represents roughly 5% of the unconditional mean of the general expense ratio (18.6%).

We next validate the compliance cost measure through a series of robustness checks. First, our results remain stable across alternative fixed-effects specifications that isolate cross-sectional differences within specific lines and years. Second, using a stacked differences-in-differences design, we demonstrate that general expenses are affected by state deregulation events, confirming that our measure captures costs directly tied to regulatory stringency. From 1992 to 2014, 22 states deregulated commercial lines. In 1992, on average, 65% of premiums in commercial lines were subject to prior approval form regulation. It dropped to 21% by 2014. Most states deregulated in the late 1990s and early 2000s. Due to the staggered timing of deregulation across states, we use a stacked differences-in-differences design (Cengiz et al., 2019; Deshpande and Li, 2019) to estimate the differences in the insurers' expenses between the years before and after deregulation.

The sample includes only insurers that operate one or more commercial lines in one state. The control group consists of insurers operating in states with prior approval regulations throughout the sample period. The treatment group comprises insurers operating in states that stopped requiring the prior approval of forms between 1995 and 2009. To address concerns that differences between insurers in the control and treatment groups may affect the estimate, we match the treatment and control groups on their regulatory environment in the year before the deregulation. We find that deregulation reduces general expenses by 1.2 percentage points, which is 8% of the unconditional mean of the expense ratio (15.2%).

We also perform a stacked differences-in-differences design for single state personal firm-line observations. Personal lines deregulation leads to an economically significant 1.8 percentage point reduction in the general expense ratio. This reduction represents a 10.6% savings relative to the unconditional mean expense ratio of 16.8%. While this evidence demonstrates that personal lines deregulation reduces the general expense ratio at the state level, we find no overall downward trend in compliance costs for personal lines nationally, as only a limited number of states have implemented personal lines deregulation.

Having established a valid measure of compliance costs associated with policy form regulation, we then examine how these costs manifest in the market. First, we observe significant economies of scale, with compliance burdens vanishing for the largest firm-line operations, while smaller firm-lines face costs exceeding the average estimates, raising the question of whether the regulation of policy forms results from regulatory capture (Stigler, 1971). Second, we find that compliance costs also differ across lines of insurance. On average, compliance costs in personal lines of insurance (e.g., private passenger auto liability or homeowners' insurance) are more than four times those in conventional commercial lines (e.g., commercial liability or special property).[3] Third, we find that compliance costs significantly decrease over the sample period for conventional commercial lines, but not for personal lines.

[3]See Appendix Table A1 and A2 and Section 2 for our classification of conventional commercial lines.

Next, we explore the non-pecuniary costs associated with policy form regulation: regulatory delays. Specifically, we investigate whether stringent oversight of policy forms is associated with longer approval times for insurers to use policy forms. To do this, we link our insurer-line-year-level financial data with detailed policy form filings gathered by S&P Global, which track the regulatory approval process of each policy form submitted by insurers in the U.S. Unlike prior research that investigates rate filings (e.g., Born et al. (2023)), this study provides a novel empirical contribution by examining form filings, which are significantly more frequent (as shown in Appendix Figure A3) and thus represent a more comprehensive measure of regulatory burden. We find that stringent form regulation is associated with approximately seven more days for final regulatory decision. Notably, the delays are longer in commercial-conventional lines than personal lines. These findings affirm that, in addition to direct financial expenditure, insurers bear significant non-pecuniary costs due to policy form regulation in the form of delayed market entry.

Lastly, we attempt to quantify the administrative paper-filing costs of insurance contract compliance. We consider regulatory compliance costs to comprise two components: legal compliance costs and administrative filing costs. Legal compliance costs are the expenses insurers incur to meet the substance of the regulation, such as ensuring the policy form itself aligns with all statutory requirements.[4] Administrative filing costs are expenses incurred to demonstrate compliance and navigate the bureaucratic process of seeking approval.[5] We exploit the exogenous adoption of the System for Electronic Rate and Form Filing (SERFF) across states to identify these costs. We hypothesize that switching to electronic filing of policy forms could reduce an insurer's administrative paper filing costs; however, it may not moderate the effect of stringent form regulation on general expenses if legal compliance costs dominate the compliance cost burden of form regulation. To address potential selection issues (i.e., insurers' decisions to select into SERFF), we leverage state mandates requiring insurers to use SERFF. SERFF was introduced in 1998, but starting in 2007, states began requiring insurers to file forms through SERFF. Our instrumental variable regression estimates suggest that SERFF usage reduces the costs of complying with form regulation by 3.7

[4]Examples of the legal compliance costs include the cost of the legal department's time to draft new, compliant contract language, the cost of the actuarial time spent ensuring rates reflect the new contract risk, or the expense of software changes needed to support a new type of required disclosure.

[5]Examples of administrative filing costs include the cost of the filing department's time to prepare and submit the form, the payment of filing fees, or the labor time spent responding to regulatory queries about the filings.

percentage points (20% of the unconditional mean expense ratio of 18.8%), but we don't find evidence that SERFF usage reduces costs differently for states subject to prior approval regulation. In summary, these results suggest that the electronic filing requirement lowers administrative filing costs, but not enough to offset the legal compliance costs.

We contribute to the burgeoning literature examining the costs of regulatory enforcement. Kalmenovitz (2023) documents the effect of paperwork regulation on corporate operating costs. Trebbi and Zhang (2022) document the relationship between regulatory intensity and labor costs, and Donelson et al. (2023) document the heterogeneity in the effectvieness of government agencies' effectiveness of regulatory enforcement. Our research setting is unique in that we leverage state-level heterogeneity in policy form regulation, which is one of the regulatory enforcement mechanisms in the insurance industry.

We also contribute to the literature by quantifying the costs of financial regulation. Most of the research has focused on bank regulation. For example, Alvero et al. (2023) and Alvero and Xiao (2020) estimate the costs of tiered bank regulation introduced with the Dodd-Frank Act of 2010. A growing body of literature evaluates the costs and benefits of financial regulations (Coates IV, 2014; Cochrane, 2014; Hahn and Tetlock, 2008; Posner and Weyl, 2013, 2014; Rose and Walker, 2020). In the insurance industry, the focus has been on studying the impact of rate regulation (e.g., Cummins and Harrington (1987); Grabowski et al. (1989); Tennyson (1993); Grace and Phillips (2008)) and the costs of complying with multiple state regulators (e.g., Grace and Klein (2000); Leverty (2012)). Policy form regulation has received far less attention. Only a few studies explore policy form regulation. Butler (2002) studies the relationship between form regulation and product innovation with data from a national commercial insurer in 1999. He finds that the time between filing forms with the state regulator and introducing a product to the market is longer under prior approval regulation. In contrast to Butler (2002), which focuses solely on commercial lines and the time to market, we estimate the costs of compliance for both commercial and personal lines. Using the 1994 deregulation of the German property and liability insurance market as a natural experiment, Berry-Stölzle and Born (2012) find that policy form regulation does not increase the unit price of insurance above competitive levels. To the best of our knowledge, this paper is the first to estimate the compliance costs of policy form regulations in the U.S. insurance market.

This study also connects with the growing scholarly literature on consumer financial contracts (Agarwal et al., 2017; Campbell et al., 2011; Houdek et al., 2018). For example, Célérier and Vallée (2017) and Jing et al. (2023) find that the complexity of retail financial products is increasing. Alexandrov (2018) studies whether laws making firms liable for the gap between consumer beliefs and contract terms reduce consumer welfare. He finds that these laws do not significantly increase prices in the credit card market, nor do they significantly decrease quantity in the mortgage market. Agarwal et al. (2015) examine the effect of regulatory limits on credit card fees and find significant savings in consumer borrowing costs. While we do not study the role of contract regulation in improving social welfare, this paper highlights the need for policymakers to consider the costs that form regulation imposes on financial institutions and consumers.

The remainder of the paper proceeds as follows. Section 2 provides the institutional background and describes the data. Section 3 presents our primary empirical strategy for measuring compliance costs and provides baseline estimates of the regulatory burden. Section 4 validates this measure through a series of robustness checks, including alternative fixed-effects specifications and an analysis of state-level deregulation events. Section 5 uses the validated measure to explore research questions regarding economies of scale, heterogeneity across lines of business, regulatory delays, and the role of electronic filing innovation in reducing costs. Section 6 concludes.

2 Form Regulation in the U.S. Property and Liability Insurance Industry

2.1 Institutional Setting

Insurance contracts are complex. They are dozens, if not hundreds, of pages long and filled with definitions, provisions, coverage extensions, additional coverages, limitations, exclusions, endorsements, and sub(limits). Many consumers lack the motivation to read the contract terms, and even if they do, they often suffer from information overload and cognitive depletion (Stark and Choplin, 2009; Becher and Unger-Aviram, 2010; Luth, 2010; OFT, 2011). As a result, all fifty states and the District of Columbia regulate the language in insurance contracts. The primary purpose of policy form regulation is to protect consumers (Tennyson, 2010). For example, Texas regulates policy forms "to ensure that the forms are not unjust, unfair, inequitable, misleading, or deceptive," [6]and Arkansas regulates forms "to establish minimum language and format standards to make property and casualty insurance policies easier to read."[7]

Insurance policies are regulated in various ways, including requiring or prohibiting specific language and terms and mandating a minimum coverage for a particular type of policy. Insurance regulators conduct form regulation through a form filing and review system set up and maintained by each state insurance department. Specifically, insurers file their contracts and any other required materials with the regulator for review and approval.[8] The state regulator (insurance commissioner) usually delegates the reviewing task to a team of reviewers, but the regulator makes the final decision. Each state also regulates insurance rates using a rate filing and review system. To date, the literature has primarily focused on rate regulation. Nevertheless, form filings account for the largest share of regulatory filings (see Appendix Figure A3).

A broad spectrum of state statutes governs the regulation of property and liability insurance contracts. The most stringent statutes require that proposed policy forms undergo a "prior approval" process, in which insurers must submit proposed policy forms to the state and obtain approval before they can use them in the market. Less stringent statutes stipulate a "file and use" system, where insurers must file forms with the state regulator (but not necessarily obtain approval) before using them. Other state laws adopt a "use and file" system, which requires insurers to file a policy form with the state regulator within a specified period after the policy is used in the market. Some statutes employ a "file only" system, which requires insurers to submit policy forms to the state without specifying a time frame. Others do not require insurers to file forms.

[6]Tex. Ins. Code § 2301.001.

[7]Ark. Code. Ann. § 23-80-302.

[8]Sometimes, the filing party is not the insurer but an advisory organization or a third-party that files on behalf of the insurer.

2.2 Data and Variables Construction

2.2.1 Stringent Form Regulation Measures

We compile a dataset on form and rate regulation stringency at the state-line level from 1992 to 2014.[9] The information on the form and rate filing systems in each state is hand-collected from the NAIC's Compendium of State Laws on Insurance Topics (1998-2014), the American Institute of Marine Underwriters (2015), and the Inland Marine Underwriters Association (2000, 2014). For the years in our sample before the 1998 NAIC Compendium of State Laws on Insurance Topics, we back-cast the statutes using information from state legislative archives and bulletins issued by state insurance commissioners.

The statutes regulating policy forms vary by line of insurance. There are 25 lines of insurance in the National Association of Insurance Commissioners (NAIC) annual statement. We categorize these lines into 14 groups according to Schedule P of the annual statement.[10] We further organize the lines into personal lines and commercial lines. The personal lines are homeowners/farmowners, private passenger auto liability, and private passenger auto physical damage. The commercial lines are special property (fire, allied lines, earthquake), commercial multiple peril, products liability, commercial auto liability, commercial auto physical damage, special liability (aircraft and boiler and machinery), other liability, workers' compensation, medical professional liability, ocean marine, and inland marine.[11] We show the categorization of the 14 lines in Appendix Table A1.

We classify a state-line-year as stringently regulated if the form filing statute in state s for line l in year t requires prior approval. Conversely, we classify a state-line-year as non-stringently regulated if the form filing statute in state s for line l in year t is file and use, use and file, file only, or no filing required. A similar binary coding rule is also used for rate regulation (i.e., prior approval equals stringent rate regulation). The dichotomous classification of form regulation into stringent or non-stringent is the most widely adopted approach in the literature (see, e.g., Harrington

[9] There are few changes to stringent form regulation after 2014. Only two states deregulated commercial forms after 2014 – Oregon in 2017 and Mississippi in 2020.

[10] Financial lines (mortgage guaranty, financial guaranty, fidelity, surety, credit, and warranty), which are considered non-traditional property-liability insurance lines (Deng, Leverty, Wunder, and Zanjani, 2025), are excluded from our analysis.

[11] Workers' compensation, medical professional liability, ocean marine, and inland marine are usually regulated under separate state statutes from other commercial lines and each other. Our regulatory stringency data reflect these variations.

(2002); Narayanamoorthy et al. (2023)). Nevertheless, the dichotomous classification may bias our estimates of the cost difference between complying with stringent and flexible regulations. First, the flexible category includes variations of regulation (e.g., file and use, use and file).[12] As a result, dichotomous categorization may bias the results against finding a difference between stringent and flexible regulation. Second, the classification of form (rate) regulation is based on how it is structured by state statutes rather than on how it is administered. In practice, state regulators can exercise discretion in administering the statute, which may result in more stringent or lenient enforcement than the statute.[13] In other words, stringent regulation is "an intention to treat" rather than a treatment, and, as such, the estimated difference between the costs of complying with stringent and flexible form regulation is biased toward zero.

The statutes regulating policy forms vary across the states and over time. From 1992 to 2014, twenty-two states deregulated their form filing system from prior approval to another type of regulation in at least one line of insurance. Appendix Tables A3 and A4 report the distribution of stringent form regulation by state and year for personal and commercial lines. For parsimony, Appendix Table A3 excludes workers' compensation, medical professional liability, ocean marine, and inland marine, as separate statutes govern these lines of insurance. Across the 23 years of our sample, 75% of the state-years are subject to stringent form regulation in personal lines, and 52% in commercial lines.

Figure 1 shows the cross-sectional variation in the stringency of form regulation laws across states at the beginning and the end of our sample period. In 1992, 39 states and the District of Columbia had at least one line of property and liability insurance regulated under a prior approval statute. Thirty-six states had forms of at least ten lines of insurance regulated by a prior approval statute. By 2014, only thirteen states had insurance policy forms of at least ten lines regulated by a prior approval statute. However, most states still regulate at least a few lines of insurance under a prior approval statute. Figure 2 shows the distribution of the number of lines subject to stringent

[12]Refer to Appendix Table A2 for a description of the differences between the major form filing systems.

[13]Anecdotal evidence on the stringency of a state's form regulation may diverge from the statutory classification. However, classifying policy form regulation using anecdotal evidence is subject to substantial measurement error, as anecdotes are subjective and there is no timing dimension to them. In contrast, statutory classification is objective and measurable over time.

form regulation for 1992 and 2014. The figure illustrates a marked decline in the number of lines requiring prior approval by 2014. The overall reduction in lines subject to the prior approval of policy forms is primarily due to the deregulation of commercial lines (see Appendix Tables A3 and A4).

We document the differences between rate and form regulation statutes in conventional commercial lines in Figure 3.[14] The figure shows whether commercial lines are regulated by a prior approval statute for rates, forms, or both rates and forms. In 1992, 17 states had prior approval statutes for both rates and forms, 21 states had form-only prior approval, 3 states had rate-only prior approval, and 10 had neither. By 2014, only two states had a prior approval statute for rates and forms, fourteen states had only forms, one had only rates, and thirty-four had neither rates nor forms. For interested readers, we also visualize the differences between rate and form statutes in personal lines in Appendix Figure A1. We again find that many states use a prior approval statute for forms but not rates.

2.2.2 Insurer Firm-Line Variables

We collect the data for all property and liability insurers in the National Association of Insurance Commissioners (NAIC) statutory annual report database for 14 lines over 23 years, from 1992 to 2014. This database is the most comprehensive source of insurer information available for the U.S. insurance market. For each year, we collect firm-line level premium and expense data from the Insurance Expense Exhibit and firm-line-state level premium, loss, and expense data from the Exhibit of Premiums and Losses ("State Page"). We use the Exhibit of Premiums Written (Schedule T) to determine an insurer's licensing status of an insurer in a state. Unlicensed insurers are exempt from form (and rate) regulations. We also collect assets, liabilities, and policyholder surplus from the balance sheet at the firm-year level.

[14] We summarize rate regulation by state for commercial and personal lines in Appendix Tables A5 and A6.

Distinct from prior studies of insurance regulation, which usually focus on a single line of business at the firm-year level, this study examines all lines of business in the property and liability insurance market and analyzes the data at the firm-line-year level. This approach is advantageous because it allows us to control for unobserved time-invariant firm and line characteristics when studying multi-line insurers.

The ideal data for this study would be insurer expenses (including salaries, consulting fees, and other expenditures) associated with regulatory compliance at the firm-line-state-year level, as the regulation of policy forms is also at this level. However, the NAIC database does not provide a separate category for regulatory compliance expenses, nor does it break down expenses at the firm-level-state-year level. We address these two challenges as follows.

First, while we do not have a single expense item dedicated to regulatory compliance, we do have two aggregate expense items: (1) Acquisition, Field Supervision, and Collection (AFSC) expenses; and (2) General Expenses. These aggregate expense items include compliance-related expenses. AFSC expenses comprise all costs incurred in producing new and renewal business, including the operating expenses of agencies and branches, the development of new policy forms, data processing, clerical and secretarial services, office maintenance, and supervisory and executive duties. General expenses include all expenses not assigned to other expense groups per NAIC statutory accounting principles. Combined, the AFSC and general expense categories encompass all expenses related to an insurer's general operations, including its costs of complying with regulations.[15] We measure expenses at the firm-line-year level using the *General Expense Ratio*, defined as the ratio of general expenses (incurred) and other AFSC expenses (incurred) to net premiums written. Second, while the treatment of stringent form regulation is at the state level, insurers do not report expenses at the state level. To address this data limitation, we measure an insurer's exposure to treatment at the firm-line-year level, the most granular NAIC data level available. *Stringent Form Proportion* is the ratio of an insurer's direct premiums written subject to stringent form regulation to total direct premiums written. In Section 3, we demonstrate that our findings are robust to the use of a different measure of exposure to stringent regulation.

[15] While the expense measure also includes expense categories not directly associated with regulatory compliance (e.g., advertising, employee welfare, rent, and equipment), our identifying assumption is that these expenses do not vary with regulatory intensity, after controlling for time-varying covariates (e.g., insurer size, insurer-by-line size) as well as unobserved firm by line and year heterogeneity through fixed effects.

2.2.3 Sample Data

The final sample is an unbalanced panel of 2,520 unique insurers and 155,793 firm-line-year observations from 1992 to 2014. In constructing the sample, we exclude firms with negative assets, negative liabilities, or policyholders' surplus of less than $1 million. At the firm-line-year level, we require observations to have positive total expenses, positive general expenses, and net premiums written of $100,000 or more.[16] We also exclude risk retention groups because they are largely exempt from regulation by non-domiciliary states (Born et al., 2009; Leverty, 2012). To reduce the effect of outliers, we winsorize the loss and expense ratios at the first and ninety-ninth percentiles.

3 Estimating the Compliance Costs of Form Regulation

In this section, we develop our empirical framework for measuring the compliance costs associated with policy form regulation. Our primary objective is to isolate the additional expenses incurred by insurers operating in "stringent" regulatory environments—defined as those requiring prior approval of policy forms—compared to those in "non-stringent" environments. Our analysis is at the firm-line level, considering each firm's line as a separate unit of business within a firm each year. Our identification strategy exploits the rich variation in the stringency of form regulation across states, lines of business, and time. First, not all states stringently regulate policy forms. Second, even in states with a prior approval system, prior approval is not required for all lines of insurance. Third, within a state-line, there is variation over time. During the study period, twenty-two states deregulate, switching from stringent to flexible form regulation in at least one line of business. Finally, multi-state insurers may conduct business in some states on a licensed basis and in other states on an unlicensed basis, and unlicensed business is exempt from form regulation.

Leveraging these variations across states, lines, and time, we estimate the following fixed effects regression model where the unit of observation is firm i's line l:

$$Y_{il,t} = \beta_1 \text{Stringent Form}_{il,t} + \gamma X_{il,t} + \lambda_{il} + \tau_t + \epsilon_{il,t} \tag{1}$$

[16] Form regulation information for Illinois in 1991 and Rhode Island from 1991 to 1997 is missing. We exclude these state-year observations from the analysis.

where $Y_{il,t}$ is the general expense ratio of firm i's line l in year t.[17] *Stringent Form* is the proportion of business written under stringent form regulation for firm i's line l in year t.[18] $X_{il,t}$ is a vector of time-varying control variables at the firm by line by year level, firm by year level, or line by year level. The control variables include the proportion of business written under stringent rate regulation statutes for firm i's line l in year t, size of the firm line (natural logarithm of net premiums written by firm i's line l in year t), size of the firm (natural logarithm of assets of firm i in year t), loss volatility (the standard deviation of the loss ratios of line l in year t), state GDP exposure of a firm i's line l in year t (gross state product of a state-year weighted by each insurer firm-line-year's direct premiums written), and state population exposure a firm i's line l in year t (total population of a state-year weighted by each insurer firm-line-year's direct premiums written);[19] λ_{il} are firm-line fixed effects that control for any unobservable firm-line-level characteristics; τ_t are year fixed effects that control for time trends. $\epsilon_{il,t}$ is a random error term. The standard errors are clustered at the firm-line-level.[20]

β_1 measures the association between the general expense ratio and the percent of premiums written by a firm-line under stringent form regulation. With the inclusion of the firm-line and year fixed effects, β_1 estimates the variation in *Stringent Form* over time (i.e., deregulation) and across states. More specifically, the firm-line fixed effects remove the variation between firms in a line, making the estimate of β_1 within the same firm and line (which only occurs when an insurer operates the line in two or more states or years with different stringencies of form regulation). The inclusion of the year fixed effects means that the estimate of β_1 also holds fixed any differences in expense usage over time. In sum, we compare how differences in exposure to stringent form regulation affect general expenses within the same firm-line, controlling for fixed differences in expense usage

[17]To ensure our findings are robust to an alternative specification of the functional form, we estimate the regression using the natural logarithm of general expenses as the dependent variable. Appendix Table A7 shows the results. We find similar results for our main variables of interest, Stringent Form and its interaction terms.

[18]One potential concern is insurers may endogenously choose the premiums they write based on the stringency of regulation. For instance, firms with lower compliance costs may gravitate to states with stringent regulation, while insurers with higher compliance costs may prefer writing in states with less regulation. We investigate the robustness of the results using alternative measures of exposure to stringent regulation. First, instead of using the proportion of premiums written in states with stringent regulation in the current year, we use premiums written in the previous year (a one-year lag variable). Second, we use the proportion of losses incurred instead of premiums written. The results using these methods are virtually the same as the main estimates.

[19]State population data come from the U.S. Census Bureau. State GDP data come from the Bureau of Economic Analysis. State GDP refers to the Gross State Product.

[20]We explore models where we cluster standard errors at the firm level and report the results in Appendix Table A8. We find consistent results.

at the year level. If β_1 is positive, expenses are higher when insurers write more premiums under stringent form regulation statutes. If it is negative, it suggests that insurers spend fewer resources under stringent form regulations. As discussed in Section 2.2.2, we dichotomize the *Stringent Form* regulation variable. Thus, the coefficient estimate of β_1 captures the relationship between "an intention to stringently regulate forms" rather than a treatment, and, as such, there is a potential attenuation bias in our estimates.

Table 1 shows the summary statistics of our sample. We denote firm-line-year level variables with the subscripts (i, l, t) and firm-year level variables with the subscripts (i, t). The sample is an unbalanced panel of insurers in 14 lines from 1992 to 2014, with 155,793 firm-line-year observations. The average firm-line writes 58% of its premiums under stringent form regulation. The distribution of the stringent form regulation variable is bi-modal, as shown in Figure 4. Approximately 20% of the observations report no premiums written under stringent form regulation. A little under 30% of the observations report 100% of premiums written under stringent form regulation. The remaining 52% of observations are approximately uniformly distributed between 0% and 100%. The average firm-line writes approximately 26% of its premiums under stringent rate regulation, although the variation across firm-lines is wide. On average, firm-lines are licensed in eleven states. The average size of a firm-line is \$44.7 million, measured by net premiums written. The average general expense ratio is 18.6%, suggesting that the average insurer spends nearly one-fifth of its net premiums written on general operations. With over \$405 billion of net premiums written in the U.S. property and liability insurance market in 2014, general expenses are large in economic magnitude (\$75.4 billion). The average loss ratio is 0.76, and the average total expense ratio is 0.34. The average insurer has assets of \$1.2 billion, and the average volatility of losses (at the line level) is 1.12. On average, State GDP exposure is \$0.473 million, and state population exposure is 11.48 million.

In Table 2, we summarize the firm-line-year level variables separately for three different categories of insurance lines: personal lines, conventional commercial lines, and other commercial lines. These line types have distinct features that would lead to differential compliance costs. Personal lines, which comprise half of the premiums written in the property and liability insurance industry (NAIC, 2018), are the most relevant lines for consumer protection. During our sample period, the deregulation of insurance forms primarily occurs in the main commercial lines, e.g., commercial multi-peril and special property. Many states govern ocean marine, inland marine, medical professional liability, and workers' compensation insurance separately from the conventional commercial insurance lines. Moreover, as shown in Table 2, Panels B and C, more observations in these lines operate in stringent form and rate regulation than other commercial lines. These lines also have longer tail risks than the other commercial lines. Accordingly, we separate these lines from the conventional commercial lines.

A majority of the firm-line-year observations belong to the conventional commercial lines (78,188 observations out of 155,793); about twenty percent of the observations belong to the "other" commercial lines (30,435 observations). On average, personal lines have the greatest premiums written in stringent form regulation at 77.2%, followed by "Commercial – Other" at 54.9% and "Commercial – Conventional" at 48.2%. The average premiums written in firm-line-years subject to stringent rate regulation is lowest for "Commercial – Conventional" lines at 16.9%. The average size of the firm-line-years, measured using net premiums written, is largest for personal lines and smallest for "Commercial – Conventional" lines. Yet, the variation in firm-line size is much greater in personal lines (coefficient of variation = 6.79) than in commercial lines (coefficient of variation = 3.95).

We report the results of estimating equation (1) without time-varying control variables in Table 3, Column (1). The coefficient on *Stringent Form* is positive and statistically significant at the 1% level. The coefficient implies that a move from 0% to 100% exposure to stringent form regulation within a given firm-line is associated with a 1.3 percentage point increase in the general expense ratio.[21] In Column (2), we add time-varying control variables. The results are robust to the inclusion of these variables, with a slightly smaller effect size. The estimated coefficient on *Stringent Form* is 0.0098 (p-value < 0.01). The magnitude represents 5% of the unconditional mean of the general expense ratio (18.6%). In monetary terms, given that the average size of a firm-line is $44.7 million in net premiums written (Table 1), the additional compliance costs associated with a firm-line moving from 0% to 100% in *Stringent Form* are approximately $438,060 per year.

[21] As shown in Figure 4, approximately 30% of the observations are writing 100% in stringent form regulation whereas 20% are not writing in stringent form regulation.

While these baseline estimates provide initial evidence of a significant compliance burden, a central challenge in measuring regulatory costs is ensuring that the observed expense differences reflect regulatory stringency rather than unobserved heterogeneity. To address these concerns and establish the credibility of our measure, we turn to a series of validation tests in the following section.

4 Validation of the Compliance Cost Measure

To ensure that our compliance cost measure accurately captures the impact of form regulation, we subject our baseline findings to a battery of validation tests. This section proceeds in two stages. First, we employ alternative fixed-effects specifications to mitigate potential biases stemming from line-specific or firm-specific time trends. Second, we move beyond cross-sectional variation to examine state-level deregulation events. By analyzing how expenses respond to the removal of stringent oversight, we provide a more direct causal test of whether our measure truly reflects the costs of compliance.

4.1 Alternative Fixed-Effects Specifications

We assess the robustness of the results to alternative fixed effects. In Table 3, we use year and firm-by-line fixed effects. In Table 4, Panel A, we replace the year fixed effects with line-by-year fixed effects to isolate the effect of cross-sectional differences in stringent form regulation.[22] The inclusion of the line-year fixed effects means that the estimate of β_1 holds fixed any differences in expense usage within a line-year. Because the firm-line fixed effects remain in the regression, β_1 is identified from deviations of a firm-line's regulatory exposure from both its own average and the line-year mean—deviations that can arise when a firm changes its geographic mix or when a state deregulates. As a result, time-series movements within a firm-line still contribute to identification, even though the comparison is cross-sectional within each line-year. In sum, the estimates of β_1 in Panel A compare how the differences in the exposure to stringent form regulation affect expense

[22] To preserve space, we only report the main coefficient of interest (*Stringent Form*).

usage within the same firm-line, controlling for fixed differences in expense usage at the line-year level. We confirm that the coefficients on *Stringent Form* (in both the baseline model and with the inclusion of the control variables) are positive and statistically different from zero at the 1% level in Panel A of Table 4.

In Table 4, Panel B, we use firm-year and line fixed effects to control for time-varying firm characteristics and unobservable line-specific differences in expense usage.[23] The firm-year fixed effects remove the variation between firms within a year, making the estimation of β_1 within the same firm and year (which occurs only when an insurer operates in two or more lines in a year with different stringency in form regulation). The inclusion of the line fixed effects, which control for constant line-level unobserved heterogeneity over time, means that the estimation of β_1 also holds fixed any differences in expense usage by line of business. In sum, the estimates of β_1 in Panel B compare how the exposure to stringent form regulation affects expense usage across different lines of business within the same firm-year, while controlling for fixed differences in expense usage at the line of business level. We show that the coefficients on *Stringent Form* are again positive and statistically different from zero at the 1% level in Panel B of Table 4. In sum, the stability of the coefficients across the various fixed effects specifications provides strong support for the validity of our compliance cost measure.

[23] Approximately 50% of the observations have no variation in *Stringent Form* within a firm-year; the median standard deviation of *Stringent Form* for a given firm-year is 0.16 and the mean is 0.18. Due to the sparse variation within a firm-year, we do not explore a model that includes both firm by year fixed effects and firm by line fixed effects.

4.2 Deregulation Tests

4.2.1 Commercial-Conventional Lines Form Deregulation

To further estimate the compliance costs of form regulation, we leverage the deregulation of policy forms. We begin our analysis with the deregulation of policy forms in conventional commercial lines, but for ease of exposition, we hereafter refer to them simply as commercial lines. During the late 1990s, there were active discussions on the excessiveness of rate and form regulation in the commercial insurance market (NAIC (1998); Eley (2000)). As shown in Appendix Table A3, many states moved away from stringent policy form regulation since 1999 (out of 22 states that deregulate commercial lines policy forms, 18 occurred between 1999 and 2009).

One challenge is that our data is at the firm-line-year level, while a line's regulation (or deregulation) is at the state-year level. Thus, multi-state insurers can contaminate the identification of the effect of deregulation as expenses at the firm-line-year level will be influenced by states with and without stringent regulation. To cleanly identify the effect of deregulation, we focus our analysis on insurers that operate commercial lines in only one state throughout the sample period (1992 to 2014). Because unlicensed business is not regulated, we only include licensed insurers.

Since the timing of policy form deregulation is staggered across the states (see Appendix Table A3), we estimate the effect of deregulation on insurer expenses using a stacked difference-in-differences approach (e.g., Cengiz et al. (2019); Deshpande and Li (2019)). We include all deregulation events with sufficient data to study a three-year deregulation window, i.e., the three years before and after the deregulation year. Since our sample period is from 1992 to 2014, the deregulation window runs from 1995 to 2011.[24] [25] Nineteen states deregulate commercial lines, in ten different years, between 1995 and 2009. These are the treated firm-lines. The control group is the firm-lines always subject to stringent form regulation throughout the sample period, from 1992 to 2014. We create ten separate panels, one for each event year. Each treatment group appears only

[24]In chronological order: Idaho (1995), Minnesota (1995), Arizona (1999), New Hampshire (1999), Pennsylvania (1999), Arkansas (2000), Louisiana (2000), Maine (2000), Virginia (2001), Alabama (2002), Michigan (2003), South Carolina (2003), Alaska (2005), Massachusetts (2005), South Dakota (2005), New Mexico (2006), West Virginia (2006), Texas (2007), and Wisconsin (2009).

[25]Within our sample period, two commercial line deregulation events occurred outside our deregulation window – New York in 2012 and Florida in 2014. While Rhode Island deregulated commercial lines in 1999, it is excluded from our sample because form regulation information is missing from 1992 to 1997.

once in each panel. The control group observations are repeated across the panels. Each panel includes the three years before and after the deregulation event year. In each panel, we only include firm-lines observed consecutively during the seven-year deregulation window (balanced panel). We then stack the panels and estimate the following model:

$$Y_{il,t} = \beta_1 \text{Deregulate}_{il,d} \times \text{Post}_{t,d} + \lambda_{il,d} + \tau_{t,d} + \epsilon_{ilt} \tag{2}$$

where d is the identifier for each of the ten deregulation panels. $\text{Deregulate}_{il,d}$ is an indicator that equals one if the firm-line is deregulated (treated) and zero otherwise; $\text{Post}_{t,d}$ is an indicator that equals one if the year is after the deregulation event year and zero otherwise. Similar to Cengiz et al. (2019) and Deshpande and Li (2019), we include within panel firm-line fixed effects ($\lambda_{i,l,d}$), and within panel year fixed effects ($\tau_{t,d}$). The standard errors are clustered at the state, the level of the deregulation treatment (Abadie et al., 2023), which we allow to be dependent across panels following Wing et al. (2024).

Our sample for the stacked differences-in-differences estimates consists of 3,899 single-state firm-line-year observations. The balanced panel consists of 557 single-state firm-lines over a 7-year period with 104 sigle-state firm-lines (728 firm-line-year observations) experiencing deregulation between the years 1995 and 2009. Summary statistics, including the univariate mean differences between the treatment and control groups, are reported in Appendix Table A9. The treatment and control groups differ in many dimensions. On average, treatment observations are less likely to operate in states with stringent rate regulation, report higher general expenses, and are smaller (both in terms of insurer total assets and total premiums written in commercial or personal lines) than the control observations.

To test the parallel trends assumption and to explore the timing of the effects, we perform an event-study analysis. We estimate the same model as equation (2), except that we include event year indicators (ranging from -3 to +3 years since the deregulation event) instead of the $\text{Post}_{t,d}$ indicator. The omitted event year is "-1", the year before deregulation. In Figure 5 Panel A, we plot the coefficients of the estimated differences in the general expense ratio between the treatment and control groups for each event year. During the pre-period, there are no statistically significant differences between the treatment and the control group, suggesting that pre-trends are not a concern. After deregulation, however, the average general expense ratio is lower for the treatment group than the control group in years 0 to +2. The estimated coefficient is approximately -0.01 over the post-event period.

We next turn to the regression results. The results from estimating equation (2) using the full sample of single-state commercial firm-line observations are shown in Table 5, Panel A, Column (1). The coefficient of interest, β_1, is the difference-in-differences estimate. It captures the average change in the general expense ratio from before to after deregulation for the treatment group relative to the change for the control group. The estimated coefficient is -0.0091, and it is not statistically different from zero.

Because rate regulation stringency can change over the event window and plausibly affect general expenses, we identify control observations that were operating in the same rate regulatory environment as the treatment observations in the year prior to the deregulation (the corresponding calendar year for the control group) and estimate a differences-in-differences model using the matched sample.[26] In Figure 5, Panel B, we perform the event-study analysis using the matched sample. While the overall trend during the pre-period is similar to that in Figure 5 Panel A, we find that the effect of deregulation during the post-period is larger in the matched sample. Using the matched sample, we find a statistically significant and large reduction in expense ratios, as shown in Table 5, Panel A, Column (2). Given that the unconditional average general expense ratio is 15.2% for the matched sample, the estimated coefficient suggests that deregulation leads to an 8% reduction in general expenses.

Since a firm (not the firm-line) may appear in the control and the treatment group in different panels, we re-estimate equation (2) using panel-specific insurer fixed effects. In Column (3), we present the results from estimating the model in Column (2) after replacing the event panel by firm-line fixed effects with event panel by insurer fixed effects. Across all specifications, we find a consistent statistically significant reduction in general expenses after deregulation.

[26] We estimate conditional treatment effects via the nearest-neighbor propensity score matched (Rosenbaum and Rubin, 1983) difference-in-differences model using a predetermined covariate following Baker et al. (2025) (e.g., Frydman and Wang (2020); Kirti and Sarin (2024)).

We also explore whether our results are sensitive to the choice of event window. Given that compliance costs arise from multiple sources, e.g., legal departments, underwriting departments, and risk management departments, the reduction in expenses can take years to materialize. Accordingly, we re-estimate the models in Table 5 Panel A using the four years (-4 and +4) since deregulation. The results are reported in Panel B of Table 5. Because the extended sample period ranges from 1992 to 2014, two deregulation events in 1995 are excluded.[27] Across the columns, we find consistent and negative coefficient estimates of deregulation. The estimated coefficients range from -0.0182 (p-value = 0.015) to -0.0217 (all significant at the 1% level). Given that the unconditional average of the general expense ratio is 15.3% for the full sample of single-state firm-lines and 14.7% for the matched sample, the estimated coefficient suggests that deregulation leads to a 11.9-14.7% reduction in general expenses. Compared to the estimated effects using the seven-year window reported in Panel A (8%), the economic effects using the nine-year window are larger, suggesting that deregulation takes time for insurers to experience a significant cost reductions.

4.2.2 Personal Lines Form Deregulation

Next, we perform similar analyses for personal lines. Relative to commercial lines, far fewer states deregulated policy forms in personal lines - only Alaska (2006), Idaho (1995), and Wisconsin (2009). We construct a dataset that consists of single-state personal firm-line observations. There are no personal firm-line observations in Alaska only. Our sample for the stacked differences-in-differences estimates for personal lines consists of 1,820 single-state firm-line-year observations. The balanced panel consists of 260 single-state firm-lines over a 7-year period, with 20 of them (140 firm-line-year observations) experiencing deregulation in Idaho and Wisconsin.

Summary statistics, including the univariate mean differences between the treatment and control groups, are reported in Appendix Table A10. We find that the treatment and control groups differ in many dimensions. Importantly, all the treatment observations operate in states that do not stringently regulate rates in personal lines, as neither Idaho nor Wisconsin required the prior approval of rate requests in personal lines. In addition, compared to control observations, the

[27] For the nine-year window sample, we identify 432 single-state firm-lines with a balanced nine-year panel (3,888 firm-line-year observations) of which 80 single-state firm-lines experience deregulation. We also perform sensitivity tests where we estimate the seven-year window excluding 1995 (i.e., the deregulation events are the same as those in the nine-year window sample) or the eight-year window and find similar effects (Appendix Table A11).

treatment observations are larger in total personal lines premiums but smaller in commercial lines premiums or in the percent of premiums written in states with stringent form regulation. Given these differences, we construct a matched sample. For each treatment observation in the year preceding deregulation, we identify a control observation that is closest to the treatment observation in terms of the rate regulatory environment, total commercial lines net premiums written, and the percentage of commercial lines net premiums written under a stringent form of regulation.

In Figure 6, we present the event-study style estimates for the full sample of the single-state personal lines observations (Panel A) and the matched sample (Panel B). We find no pre-trends in either graph. Although we do not find statistically significant differences in expenses during the post-period in the full sample, we do find a significant reduction in the matched sample. We report the results of the differences-in-differences estimates in Table 6, Panel A. With the matched sample, we find that personal lines form deregulation results in a 10.6% reduction in general expenses with a coefficient estimate of -0.0178 (-0.0178/0.163, p-value = 0.054). The result is robust to alternative combinations of fixed effects (Columns (3)-(5)) as well as widening the event window to include +4 year since deregulation, with a slightly reduced coefficient estimate of -0.0153 (Panel B).[28]

In summary, we find that form deregulation results in an average reduction in general expenses of approximately 8-10% for both commercial lines and personal lines. Since the general expense ratio is expressed as a percentage of net premiums written, the estimated reduction in the general expense ratio following deregulation can be translated into the expense-loading component of insurance premiums without additional pass-through assumptions (Actuarial Standards Board, 1997; Werner and Modlin, 2016). Using the coefficient estimates on the reductions in general expenses relative to premiums following deregulation, the economic impact corresponds to $12 to $22 in savings per $1,000 of premiums for commercial lines and $15-$18 for personal lines. The estimated economic impact is larger than our correlation estimates in Section 3.[29]

[28] Because Idaho's personal lines form deregulation occurred in 1995, we do not estimate models with a wider pre-event window.

[29] Two factors likely account for the gap. First, the deregulation tests are restricted to single-state insurers, which are on average smaller than the multi-state insurers in the Section 3 sample (Appendix Table A9 and A10) and likely face proportionally higher fixed compliance costs per dollar of premium. Second, the differences-in-differences design isolates within-firm changes around regulatory events and absorbs time-invariant heterogeneity that the pooled OLS specification in Section 3 cannot, leading to an estimate that more cleanly reflects the marginal cost of prior approval rather than an average cross-sectional association.

5 Applications of the Compliance Cost Measure

The stability of the coefficients across various specifications and the significant reductions in expenses following deregulation events provide strong support for the validity of our compliance cost measure. Having established that our measure reliably identifies the regulatory burden, we now apply it to examine how costs vary across market segments and to investigate the broader economic implications for the insurance industry. This section applies our empirical framework to four key areas of inquiry. First, we explore whether there are economies of scale in regulatory compliance. Second, we examine whether there is heterogeneity in costs across personal and commercial lines and over time. Third, we investigate whether there are non-pecuniary costs associated with regulatory delays. Finally, we use our measure to assess the efficacy of the SERFF electronic filing innovation in reducing administrative expenses.

5.1 Economies of Scale in Compliance

We next explore whether there are economies of scale in complying with stringent policy form regulation. Specifically, we augment the regression model in Table 3 Column (2) by interacting *Stringent Form* with the natural logarithm of the firm-line size (ln(Firm-Line Size)). Figure 7 plots the marginal effect of stringent form regulation as a function of firm-line size, holding all other variables constant. The cost of complying with stringent form regulation decreases as firm-line size increases. Across the distribution of firm-line size (from smallest to largest), the marginal effect of stringent form regulation is 2.1 percentage points at the 10th percentile (firm-line size = \$0.4 million); 1.4 percentage points at the 25th percentile (firm-line size = \$1.23 million); and 0.06 at the 50th percentile (firm-line size = \$4.84 million). Stringent form regulation does not increase the compliance costs of the largest firm-lines, as the marginal effect of stringent form regulation is not statistically different from zero at or above the median firm-line size. In sum, the burden of complying with stringent form regulations falls unevenly on firms, with the smallest firm-lines being greatly impacted while the largest firm-lines can amortize compliance costs across their many policyholders. The evidence suggests that stringent form regulation may have a deleterious effect on competition, as small entrants are at a competitive disadvantage relative to large incumbent firms. It is also suggestive of regulatory capture, as larger insurers may desire stringent form regulation because it provides them with a competitive advantage relative to small insurers.

5.2 Heterogeneity Across Lines of Business and Over Time

To investigate whether the impact of stringent form regulation differs by line type, we re-estimate equation (1) separately for personal lines, conventional commercial lines, and other commercial lines. The regression results are shown in Table 7, Columns (1), (2), and (3), respectively. The coefficient estimate on *Stringent Form* is positive and statistically significant at the 10% level for all three line types; however, its magnitude is smaller for conventional commercial lines (CM) relative to personal lines (PS) and other commercial lines (CM-Other). The estimated coefficient for personal lines (0.026, p-value < 0.01) explains 14.8% of the unconditional average of the general expense ratio for personal lines (Table 2, Panel A), while the estimated coefficient for the conventional commercial lines (0.006, p-value $= 0.07$) explains only 3.1% of the unconditional average general expense ratio (Table 2, Panel B). In summary, the burden of complying with policy form regulation is larger for personal lines than commercial lines.

We next explore whether the compliance costs associated with form regulation vary over time.[30] We re-estimate the models in Table 3 Column (2) along with those in Table 7 Columns (1) to (3), including interactions of the year fixed effects with *Stringent Form*. We then calculate the marginal effect of *Stringent Form* on the general expense ratio by year and line type. The results are shown in Figure 8. For all the lines combined (Panel A), the marginal effect is relatively stable from 1992 to 2008 but then drops in 2009. The marginal effect averages 0.020 from 1992 to 2008, but it only averages 0.0017 from 2009 to 2014. Across the sample period, the marginal effect for personal lines

[30] Appendix Figure A2 shows that the proportion of premiums subject to stringent form regulation falls from 1992 to 2014. Panel A shows that 68 percent of premiums were subject to stringent form regulation in 1992, but only 41 percent in 2014. Panels B-D show that the drop was primarily in the conventional commercial lines (Panel B). The proportion of conventional commercial lines with stringent form regulation was approximately 65 percent in 1992 but only 20 percent in 2014. Panel C shows the proportion was relatively flat in personal lines, while Panel D shows a relatively modest decline in other commercial lines.

(Panel B) averages 0.032, with a low of 0.018 in 1992 and a high of 0.057 in 2008. There is a general increasing trend in compliance costs in personal lines. The marginal effect for conventional commercial lines (Panel C) averages 0.007 across the sample period, with an average of 0.019 from 1992 to 2007 and an average of -0.022 from 2008 to 2014.

5.3 Non-pecuniary Costs (Regulatory Delays)

In this section, we explore whether stringent form regulation affects how long it takes regulators to approve policy forms. Using one U.S. commercial insurer in 1999, Butler (2002) finds that the form regulation is positively associated with delays in product innovation. Our analysis uses a novel dataset that includes form filings from insurers at the line and state levels, sourced from the Standard & Poor's Market Intelligence Property & Casualty Rate and Product Filings data. The data includes the NAIC company code, business lines, insurance line sub-type, regulatory decision, submission date, and disposition date for each form filing submission for years 2007 to 2014.[31] We include form filings as well as filings that combine form, rate, and rule filings.

To cleanly identify the effect of stringent form regulation on regulatory delays, we aggregate the form filing submissions at the insurer-line-state-year-level for the years 2007 to 2014.[32] For each insurer-line-state-year unit, we then calculate the mean number of days from the date of submission of a form filing to the date of disposition (to limit the influence of outliers, we winsorize the variable at the 1st and 99th percentiles).[33] To explore regulatory delays for our sample insurers, we link the dataset with the sample NAIC data, which includes insurer financial variables.[34]

[31] Standard & Poor's Market Intelligence notes 2007 as the year that most states report product and rate filings comprehensively (S&P Global Market Intelligence, 2025). These states have comprehensive coverage after 2007: Colorado (2008), New York (2010), Hawaii (2013), New Mexico (2014), and Louisiana (2015). Our data is similar to the rate filings data utilized by Born et al. (2023), except we include all business lines and only include form filings.

[32] Our results are similar when we use the individual form filings, rather than those aggregated at the insurer-line-state-year-level.

[33] Approximately 94.5% of the form submissions are approved. Among the remaining 5.5%, 2.5% are disapproved and 3% are withdrawn by the insurance firms.

[34] The lines of business reported in the Standard & Poor's Market Intelligence Property & Casualty Rate and Product Filings data match with the business lines identified in insurers' NAIC statutory statements, except for auto lines. Private passenger auto liability and private passenger auto physical damage are reported as "personal auto". The same goes for commercial auto. We, therefore, combine auto liability and auto physical damage into one auto line (for both personal and commercial auto). As a result, our analysis of the Standard & Poor's Market Intelligence Property & Casualty Rate and Product Filings data uses 12 lines of business to perform analysis. We provide the summary statistics and univariate differences between the stringent form regulation environment for the Rate and Product Filings data in Appendix Table A12.

We estimate a model that includes firm-by-line and year fixed effects for the firm, line, state, year observations:

$$Y_{ils,t} = \beta_1 I(\text{Stringent Form})_{ls,t} + \gamma X_{ils,t} + \lambda_{il} + \tau_t + \epsilon_{ils,t} \tag{3}$$

where $Y_{ils,t}$ is the average days since submission to regulatory disposition of firm i's line l form submission in state s in year t. $I(\text{Stringent Form})_{ls,t}$ is an indicator variable that equals one if state (s) regulates line (l) under stringent form regulation in year t, and zero otherwise. Because this indicator is defined at the state-line-year level, it takes the same value for all firms i submitting forms in that state-line-year. We include a vector of control variables, $X_{ils,t}$, which include firm-line-state-level, firm-line-level, firm-level, line-state-level, and state-level time-varying characteristics: size of firm i's line l in state s (natural logarithm of direct premiums written by firm i's line l in state s in year t), size of firm i's line l (natural logarithm of net premiums written by firm i's line l in year t), size of firm i (natural logarithm of assets of firm i in year t), an indicator that state s requires prior approval of rates for line l in year t, the natural logarithm of the total number of form submissions received by state s for line l in year t, the natural logarithm of the state's GDP (i.e., gross state product), and the natural logarithm of the state population;[35] λ_{il} are firm-line fixed effects; and τ_t are year fixed effects. $\epsilon_{ils,t}$ is a random error term. The standard errors are clustered at the state, the level of stringent form assignment (Abadie et al., 2023).

β_1 measures the association between the time to policy form disposition and stringent form regulation. The firm-line fixed effects remove the variation between firms in a line, making the estimate of β_1 within the same firm and line (which only occurs when an insurer operates the line in two or more states or years with and without stringent form regulation). The inclusion of the year fixed effects means that the estimate of β_1 also holds fixed any differences in the time to disposition over time. In sum, we compare how differences in exposure to stringent form regulation affect the regulatory time to disposition within the same firm-line, controlling for fixed differences in the time to disposition at the state and year level. As a result, a positive β_1 implies a longer time to disposition due to variation in stringent form regulation across states each year, as well as within states over time.

[35]State GDP and state population variables in the equation differs from the state exposure variables in equation (1), because our unit of observation is at the firm-line-state-year.

The results from estimating equation (3) are reported in Table 8. Column (1) shows the results without time-varying control variables. The coefficient on *Stringent Form* is positive, but it is not statistically different than zero. After including time-varying control variables, Column (2), we find that stringent form regulation delays the disposition of forms by approximately one week (7.71, p-value = 0.05). Columns (3) to (5) analyze the line type subsamples. The coefficient on *Stringent Form* is statistically different from zero only for the commercial-conventional lines sample, Column (4). The coefficient implies that stringent form regulation delays disposition of forms by approximately ten days (9.8, p-value = 0.046). While modest in terms of the number of days, the effect size is economically meaningful relative to the average number of days until a regulatory decision. The estimated effect size is approximately 38% of the average number of days until a regulatory decision.[36]

To isolate the effects of changes in stringent form regulation over time, we estimate a model that replaces the firm by line fixed effects with firm by line by state fixed effects. Because the year fixed effects remain, the fixed effects absorb variation across line-states within a given year and capture only the differences in days until disposition for states that relax stringent form regulation over time.[37] The results are reported in Table 9. Without the time-varying control variables (Column (1)), we do not find an effect of deregulation on the time until disposition for the full sample. With the time-varying controls (Column 2), we find that stringent form regulation leads to a statistically significant delay in the time to disposition. The analyses of the line subsamples (Columns (3) to (5)) show that the delays exist in both personal and commercial-conventional lines. The magnitude of the delay is greater in commercial-conventional lines (column (4)). The effect size in column (4) is also larger than the estimates reported in Table 8 column (4), strengthening the inference that prior approval lengthens disposition time.

[36] We find similar results when we replace the year fixed effects with: i) line by year fixed effects; or ii) firm by year fixed effects.

[37] Unfortunately, data limitations prevent us from estimating the causal effect of stringent form deregulation on the number of days until disposition using a differences-in-differences approach similar to Section **??**. There are three states that de-regulate commercial forms since 2007: Texas (2007), Wisconsin (2009), and New York (2012). The Rate and Product Filings data only comprehensively records the dates until disposition starting in 2007. Because the difference-in-differences design requires a balanced +/- 3-year window, Texas lacks a pre-period and New York lacks a full three-year post-period. Only Wisconsin satisfies the event-window criterion.

5.4 SERFF Innovation

In this section, we examine an innovation in the property and liability insurance industry that, in theory, reduces administrative filing costs associated with compliance, but should not materially impact the legal costs of compliance.[38] Before filing policies with state regulators, insurers hire staff, consultants, or lawyers to prepare the contracts and materials required by the state. Insurers also incur costs when modifying policies to accommodate market demand or when managing the legal risks posed by recent court decisions.[39] These are legal compliance costs, and the central component of policy form regulation to promote consumer protection. Yet, policy form regulation also creates administrative filing costs that can be burdensome to insurance companies. For example, at the filing and review stage, insurers pay filing fees and spend time and resources corresponding with reviewers when contract modifications are necessary. The costs can recur annually, as policy forms are required when insurers introduce a new policy in a line and state, and even multiple times within a year if insurers modify policy forms.

To investigate whether innovation can lower regulatory compliance costs, we leverage a regulatory intervention aimed at reducing administrative filing costs. To improve the efficiency of the regulatory filing process, state regulators and the NAIC started a "Speed to Market for Insurance Products" initiative in the mid-1990s. This initiative led to the establishment of the System for Electronic Rate and Form Filing (SERFF) in 1998. SERFF is an internet platform designed to streamline the filing and review process of rates and policy forms. The NAIC has promoted SERFF as a major "Speed to Market" tool since its first product release. When the first version of SERFF was released in 1998, eight states and sixteen insurers participated. A total of 294 filings were made using SERFF in 1998. In 2000, an enhanced version of SERFF was released, and it quickly expanded across the states in the early 2000s. By the end of 2004, SERFF was accepted in 49 states and the District of Columbia.

[38] Form regulation can also have implicit costs by disadvantaging insurers when competing with alternative risk transfer mechanisms (Butler, 2002).

[39] Many insurers have task forces specializing in policy form compliance.

SERFF replaces the traditional paper-based system by digitizing the entire filing process. Filers log onto SERFF, identify the filing requirements promulgated by each state, and submit an electronic filing to the regulator, accompanied by all supporting documents uploaded in digital format. The regulator receives the filing on SERFF and can start reviewing it instantly. In cases where a change is needed, the regulator can contact the filing insurer via the SERFF messaging system, and the insurer can make the necessary changes and submit a revised filing in SERFF. In short, SERFF provides a one-stop interface for regulators and insurers to exchange information and complete rate and form filings electronically. Importantly, SERFF only alters how rates and forms are filed and processed; it does not modify the legal compliance process.

SERFF has several advantages over the traditional filing system, particularly in terms of administrative filing costs. First, SERFF improves the efficiency of the information exchange by providing a single online platform for insurers and regulators to communicate, manage, and store policy form and rate filings electronically. Insurers no longer need to mail hard copies of product information to the regulator and wait for a response in the mail. Second, SERFF makes the administrative filing process more manageable by providing insurers with easy access to the state's current filing requirements promulgated by the state, which helps insurers submit more accurate and complete filings. Moreover, SERFF is particularly helpful to multi-state insurers, as they can comply with the regulations across different states much more easily and even use a single filing if it meets the regulatory requirements of multiple states.[40] Industry reports suggest that the adoption of SERFF leads to considerable cost savings, including reductions in internal IT and database maintenance costs, as well as product filing worker hours (NAIC, 2016).

Insurers, however, incur both fixed and variable costs when transitioning from the traditional filing method to SERFF. For an insurer to implement SERFF, it needs to allocate resources to staff training, and potentially upgrade its technology. Also, insurers pay a filing fee for each filing they submit, either on a pay-as-you-go basis or by purchasing for a "block" of filings at a lower price in advance.[41] Insurers vary in when (and to what extent) they transition to SERFF.

[40] There may be spillover reduction in legal compliance works for multi-state insurers, yet we view this to be limited. For example, in the life insurance industry, the usage of SERFF is directly linked with a product standardization process, the Interstate Compact, which is expected to reduce legal compliance costs. In the property and liability insurance industry, however, SERFF does not lead to such a product standardization.

[41] As of 2024, the rate is \$18.68 per filing. In the previous years, a prepaid rate was offered at lower rates. For example, in 2019, the prepaid rates were \$9.50/\$8.00/\$6.50 per filing for a block of 500/1,000/1,500 filings.

We aim to quantify the administrative filing costs of complying with form regulations using SERFF. Our results in Section 3 provide evidence that insurers incur substantial costs to comply with stringent form regulations. When it comes to the impact of SERFF on the costs of complying with prior-approval form regulations, it is not obvious whether SERFF adoption would reduce the positive association between *Stringent Form* and *General Expense Ratio*, or whether it would have no moderating effect. While we hypothesize that SERFF will reduce administrative filing costs, it is unclear whether administrative filing costs or legal compliance costs dominate the regulatory compliance costs of prior approval form regulation. First, if administrative filing costs account for a significant portion of total regulatory compliance costs (i.e., administrative filing costs are significantly higher in regulated states than in non-regulated states), then SERFF can attenuate the positive association between *Stringent Form* and *General Expense Ratio.* Second, if administrative filing costs are not material and regulatory compliance costs are dominated by legal compliance costs, then SERFF adoption would have little or no effect on the positive association between Stringent Form and General Expense Ratio.

A key concern with using SERFF usage to identify the administrative filing costs of form regulation is that insurers' selection into SERFF use may bias our estimates in an ordinary least squares setting.[42] To evaluate whether such selection bias is a concern, we create a summary measure of SERFF usage at the firm-line-year-level using the SERFF form filings data compiled by the NAIC for the 1992-2014 period. We measure SERFF usage only for the filing of policy forms.[43] For each firm-line in a year and state, we observe whether the insurer utilizes SERFF to file forms. We then merge it with the insurer-line-state-level premiums to calculate the proportion of premiums written using SERFF. In Figure 9, Panel A, we show the distribution of SERFF usage during the 2007-2014 period, which follows the first state mandates to use SERFF in Georgia and South Dakota in 2007. We find that over 60 percent of the observations do not use SERFF, suggesting that the level of SERFF usage differs across firm-lines. In Figure 9, Panel B, we show that the average SERFF usage

[42]In particular, insurer expenses depend on how insurers file their policy forms with regulators. As such, we can imagine a second equation in which insurers choose how to file policy forms based on the filing costs. In this case, the ordinary least squares estimates using a single equation will be inconsistent.

[43]We find that of the total number of P&L insurer filings with SERFF, approximately 45-50% are form only filings, and less than 10% are rate only filings, as shown in Appendix Figure A3. Insurers can also submit a combined form filings (e.g., form and rate; form and rule; form, rate, and rule), representing approximately 13% of SERFF filings. In constructing our SERFF usage variable, we include form-only and combined form filings.

proportion has been increasing since 2003, with a steep increase after 2006. We see another jump in usage from 2009 to 2010, which coincides with multiple states mandating SERFF in 2008 and 2009. The figures reveal that selection is a concern, yet they also highlight a positive correlation between SERFF mandates and SERFF usage.

To address the potential endogeneity, we use instrumental variables estimation. We use the year that the SERFF system becomes mandatory in a state as the instrumental variable (IV) for SERFF usage. Under a state's SERFF mandate, insurers can no longer make paper filings in the state and must file proposed forms using SERFF; therefore, a state requiring SERFF is highly correlated with the usage of SERFF satisfying the IV relevance condition. The IV satisfies the exclusion condition because the mandate is exogenous. The IV also satisfies the exclusion condition as the SERFF mandate affects expenses only through the switch to electronic filing, and the mandate's timing is unlikely to coincide with other state-level cost initiatives (after including firm-line and year fixed effects). The mandate itself does not lower insurer expenses, but it can affect expenses through the usage of the SERFF system. In other words, our IV strategy identifies the effect of SERFF on the administrative costs of compliance for the insurers who are induced to file forms electronically due to the SERFF mandate.

Many states have mandated the use of SERFF for insurance form filings since 2007. We obtain information on which states mandate SERFF and the effective dates of the mandate from the official NAIC-SERFF website.[44] We summarize the staggered state-level SERFF mandate in Figure 10. Between 2007 and 2014, 31 states enacted SERFF mandates for property and liability insurance. Florida is the only state that does not accept SERFF, as it adopted its own electronic filing system, named I-File (later replaced by the Insurance Regulation Filing System (IRFS)), and mandated its use in 2007.[45] We conservatively categorize Florida as a non-SERFF state.[46] Additionally, multi-state insurers can conduct business in some states on a licensed basis and in other states on an unlicensed basis; only licensed business is subject to policy form regulation. As a result, an unlicensed business does not involve using SERFF at all, providing additional variation in SERFF

[44] https://www.serff.com/documents/serff participation mandates.pdf.

[45] Florida also required electronic filings beginning July 2003 either through I-File or a diskette.

[46] While I-File is similar to SERFF in that in replaces paper filings, it is unique to Florida so there is no multi-state advantage as with SERFF. We, therefore, categorize Florida as a non-SERFF state.

usage. To be consistent with our annual insurer data, we classify the SERFF mandate at the year level; if the mandate occurs after July 1st of year t, we define the mandate's first year to be t+1. The classification of SERFF mandates at the year level may introduce measurement error, which would bias us against finding any significant SERFF effect on insurers.

Our two-stage instrumental variable (IV) estimation model is:

$$I(\text{SERFF Usage})_{il,t} = \beta \text{SERFF Mandate}_{il,t} + \beta_1 \text{Stringent Form}_{il,t} + \gamma X_{il,t} + \lambda_{il} + \tau_t + \epsilon_{il,t} \quad (4)$$

$$Y_{il,t} = \beta_{IV} I(\widehat{\text{SERFF Usage}})_{il,t} + \beta_1 \text{Stringent Form}_{il,t} + \gamma X_{il,t} + \lambda_{il} + \tau_t + \epsilon_{il,t} \quad (5)$$

where $Y_{il,t}$ is the general expense ratio of firm i's line l in year t.[47] $I(\text{SERFF Usage})_{il,t}$ is an indicator variable of whether firm i uses SERFF in line l and year t;[48] and SERFF Mandate$_{il,t}$ is an indicator of whether firm i's line l is subject to a *SERFF Mandate*. If at least one of the states in which firm i writes line l (and is licensed) is subject to a *SERFF Mandate*, we consider the firm-line to be subject to the mandate. In the first stage, we estimate *SERFF Usage* using *SERFF Mandate* as an instrument. Predicted *SERFF Usage*, $I(\widehat{\text{SERFF Usage}})_{il,t}$, is then used in the second stage to estimate the association between SERFF usage and general expenses. All other variables are defined in equation (1). The standard errors are clustered at the firm-line level.

Motivated by the scarce usage of SERFF before 2002, we estimate the regression using data from 2002 to 2014. We report the results of the two-stage instrumental variable regression estimation in Table 10. Because instrumental variables identify the effect on the outcome variable for those induced to switch due to the instrument, in Table 10, we estimate the effect on insurer expenses for insurers that are induced to switch to electronic filing because of state mandates. In Column (1), the coefficient estimate on $I(\text{SERFF Usage})$ is -0.037 and statistically different from zero (p-value < 0.01). The coefficient estimate suggests that insurers using SERFF report general expense ratios that are 3.7 percentage points lower than those not using SERFF. Thus, the results confirm that electronic filing reduces the general expenses associated with form regulation. Given that the

[47]We do not explore regulatory delays as the outcome variable. Because SERFF adoption is nearly universal—about 95% of filings are submitted through SERFF—we lack a sufficient number of non-SERFF observations to credibly estimate SERFF's effect on regulatory delays.

[48]Due to the bimodal distribution of insurers' SERFF usage as shown in Figure 9, we use an indicator variable of SERFF usage.

unconditional general expense ratio is 18.8 percent, the coefficient estimate implies that SERFF usage reduces the general expense ratio by 19.5%. The associated F-statistic for the regression is 956.4, well above the critical value of the weak instrument test (16.38) based on Stock and Yogo (2005). The first stage results of the model are in Appendix Table A13. The estimated coefficient on I(SERFF Usage) is 0.197 and statistically different from zero (p-value < 0.01), suggesting that firm-lines subject to the mandate are approximately 20% more likely to use SERFF.

In column (2), we interact stringent form regulation with SERFF usage. The coefficient estimate on I(SERFF Usage) is -0.043 and statistically different from zero (p-value < 0.01). The coefficient on the interaction of *Stringent Form* and I(SERFF Usage) is not statistically different than zero, supporting the hypothesis that legal compliance costs dominate the compliance costs of stringent form regulation. Overall, the results in column (2) suggest that, on average, SERFF reduces the administrative filing costs of form regulation but likely does not offset the legal compliance costs of stringent form regulation.

Lastly, we explore whether the results are sensitive to allowing correlations within insurers. SERFF mandates are not specific to the line but apply to all business lines operating in a given state. Our instrument, therefore, may vary across insurers and not across firm-lines. If our two-stage least squares models do not accurately capture the firm-line-level variation in compliance costs arising from SERFF, our results would be different if we cluster the standard errors at the insurer level. In Columns (3) and (4) of Table 10, we explore this hypothesis by clustering standard errors at the insurer level. The results, despite modestly larger standard errors, are consistent with the results reported in Columns (1) and (2), confirming that the effect of SERFF on firm-lines is robust to allowing correlations within insurers.

6 Conclusion

This study provides the first evidence of the compliance costs associated with form regulation in the U.S. property-liability insurance industry. We contribute to the empirical literature on financial industry regulation and on regulation in the economy more generally. We analyze a panel data set of the insurers covering all lines of property-liability insurance and find significantly higher expenses associated with stringent form regulation. The costs are greater for small-firm lines of operations and for personal lines of insurance. We also find that innovations in regulatory efficiency can bring meaningful declines in the costs of regulatory compliance. Specifically, we find that SERFF, an electronic filing system, reduces the administrative filing costs of form regulation, but not large enough to offset the legal compliance costs of form regulation.

To overcome the concern that insurers can choose to operate in states and lines with favorable form regulation, we estimate the effect of form regulation on compliance costs by leveraging the deregulation of form regulations. We use stacked difference-in-differences regressions with the subsample of single-state firm-lines to establish causality and find that deregulation reduces general expenses, with a meaningful economic impact in both commercial and personal lines. Since the general expense ratio is expressed as a percentage of premiums, the estimated reduction following deregulation can be translated into reductions in the expense loading component of insurance premiums without additional pass-through assumptions. For commercial lines, the estimated reduction in the expense ratio translates to \$12-\$22 in savings per \$1,000 in premiums. For personal lines, the savings are \$15-\$18 per \$1,000 premiums. The personal lines estimates are particularly policy-relevant given current concerns about the affordability of homeowners and auto insurance.[49] The implications of these estimates vary by insurer size and geographic scope. The single-state sample utilized in the policy form deregulation study is disproportionately composed of small-firm lines, for which compliance costs are higher per dollar of premium, suggesting these estimates best represent the compliance costs for smaller local insurers. For large multi-state insurers, the ability to amortize fixed compliance costs across many states means the per-premium compliance cost savings from deregulation could be small, consistent with the documented economies of scale. Nevertheless, the aggregate dollar savings for larger insurers would be larger than those for small insurers, given the premium volumes. Moreover, the cumulative compliance burdens from operating across heterogeneous regulatory environments suggest they may pass through to consumers more than our single-state estimates imply.

[49] Availability is a second-order market outcome influenced by multiple confounding variables (macroeconomic trends, catastrophes, reinsurance pricing, etc.), whereas insurer expenses are a first-order, direct consequence of the regulatory filing process itself. As such, the primary mechanism through which policy form regulation operates is compliance costs. Moving from expenses to availability requires a series of assumptions about the elasticity of supply and demand and the pass-through costs.

We use our compliance cost measure to investigate how compliance costs vary over time and across lines of business, whether there are economies of scale, and whether electronic filing innovations reduce compliance costs. Future research can build upon the measure introduced in this paper to explore other research questions, for example whether and how stringent policy form regulation changes the structure of the market, such as the level of competition (including entry and exit decisions) or the tension between the admitted and the non-admitted insurers (surplus lines or risk retention groups). While we document that costs of complying with stringent form regulation are economically meaningful, they may not be unwarranted given the potential benefits of the extra regulatory scrutiny, including consumer protection and higher insurance demand (Butler, 2002). We look forward to future research on the benefits of form regulation, which, combined with this study, can help provide a cost-benefit comparison.

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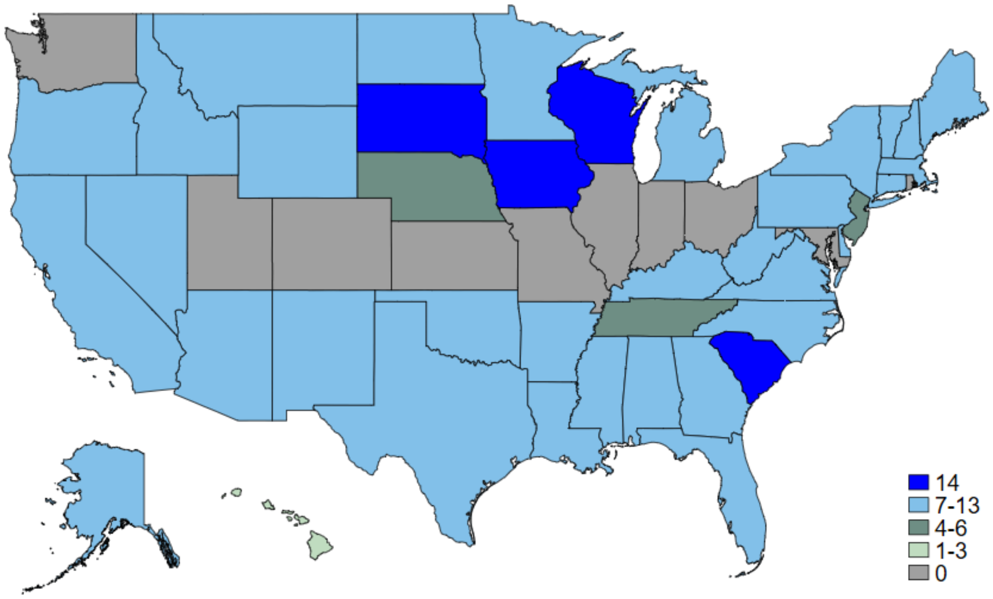
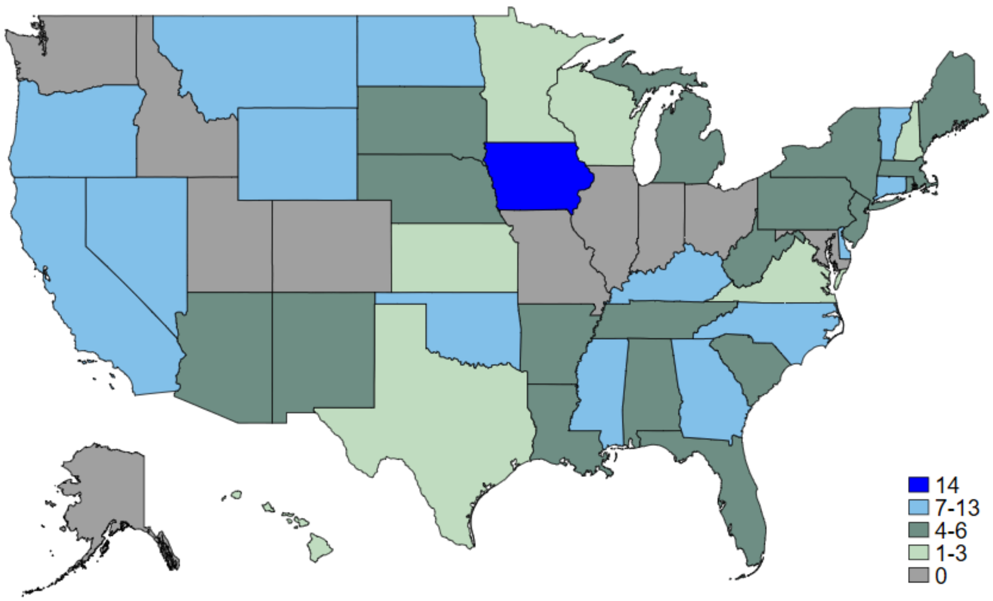
Tables and Figures

Figure 1: Number of Lines under Stringent Form Regulation

(A) 1992

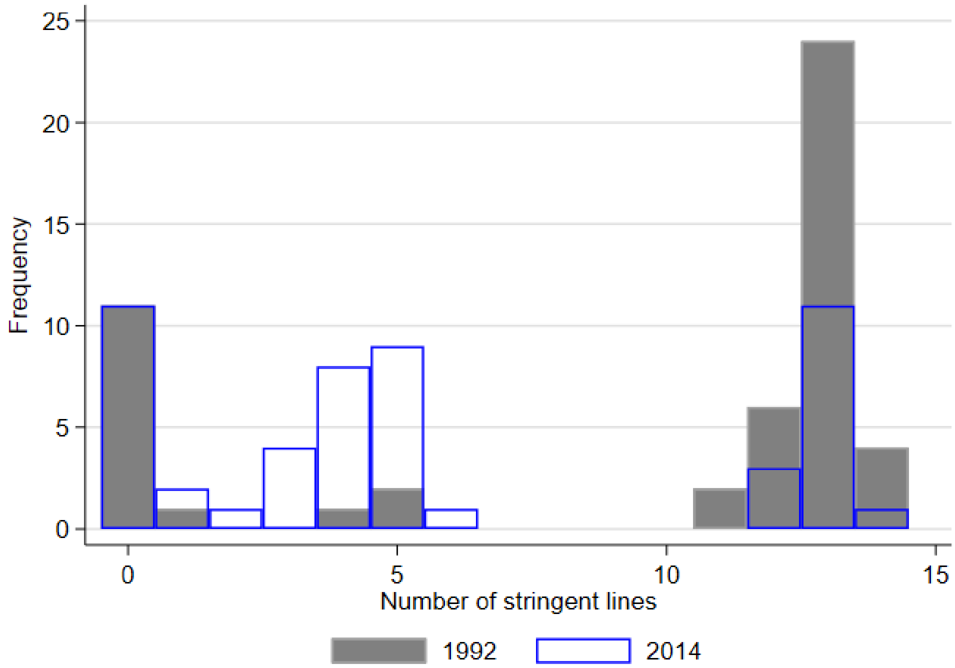


(B) 2014



Notes: The figures show the number of lines under stringent form regulation (i.e., "Prior Approval") by state for the years 1992 (panel (A)) and 2014 (panel (B)). See Appendix Table A1 for the definitions of business lines and Appendix Table A2 for a description of the types of form regulation. Data sources: NAIC (1992, 2014) and state statutes.

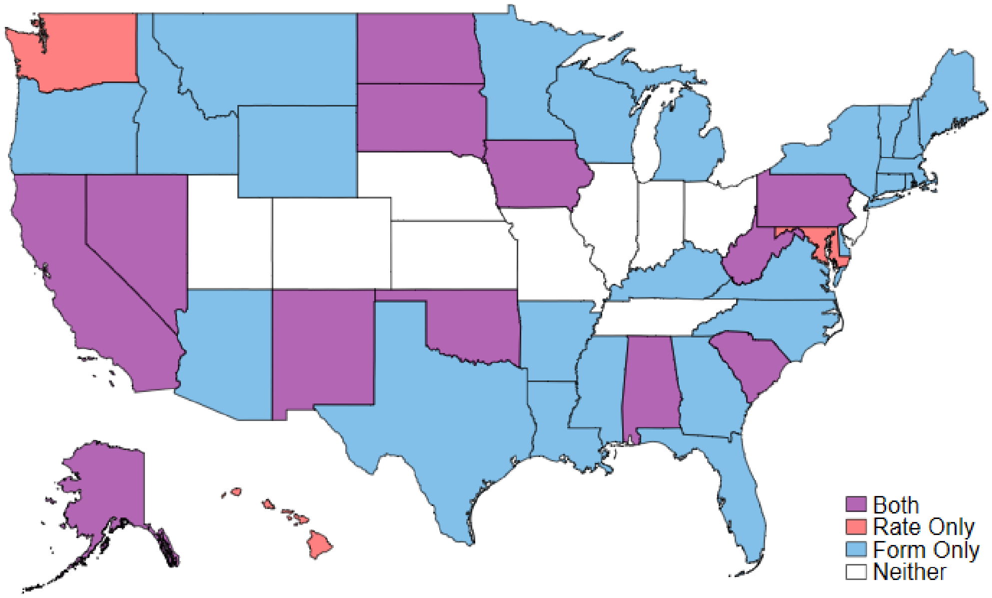
Figure 2: Distributions of the Number of Lines under Stringent Form Regulation by Year



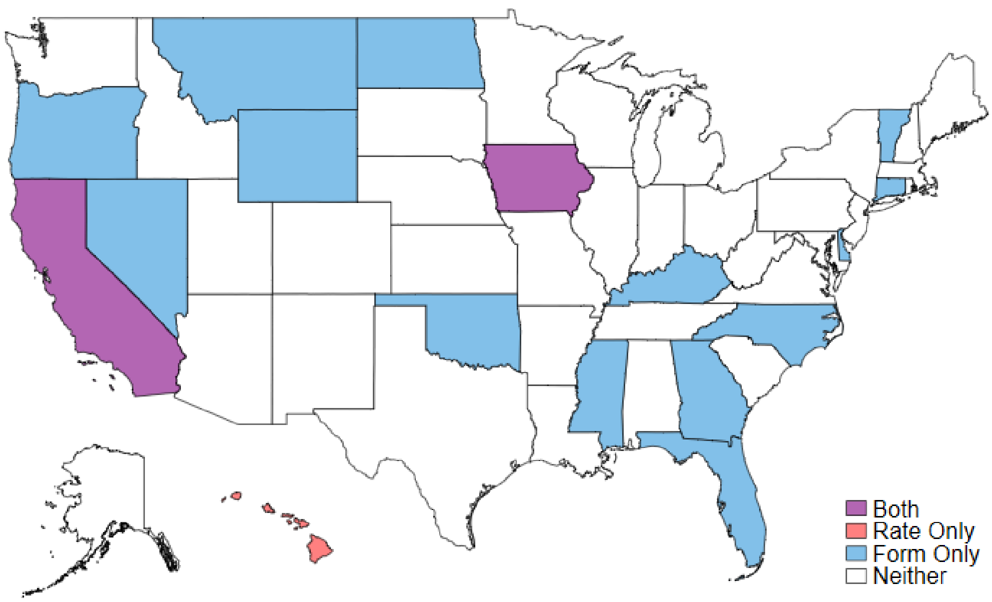
Notes: The histogram show the distribution of the number of lines under stringent form regulation (i.e., "Prior Approval") across states for the years 1992 and 2014. See Appendix Table A1 for the definitions of business lines and Appendix Table A2 for a description of the types of form regulation.
Data sources: NAIC (1992, 2014) and state statutes.

Figure 3: Stringent Form X Rate Regulation, Commercial

(A) Commercial Lines in 1992

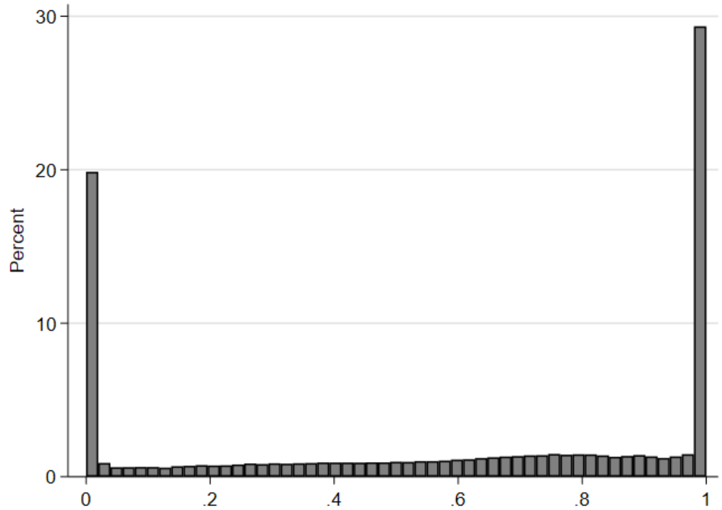


(B) Commercial Lines in 2014



Notes: The figures show whether forms and/or rates of the conventional commercial lines are subject to "Prior Approval" in 1992 (panel (A)) and 2014 (panel (B)).
Data sources: NAIC (1992, 2014) and state statutes.

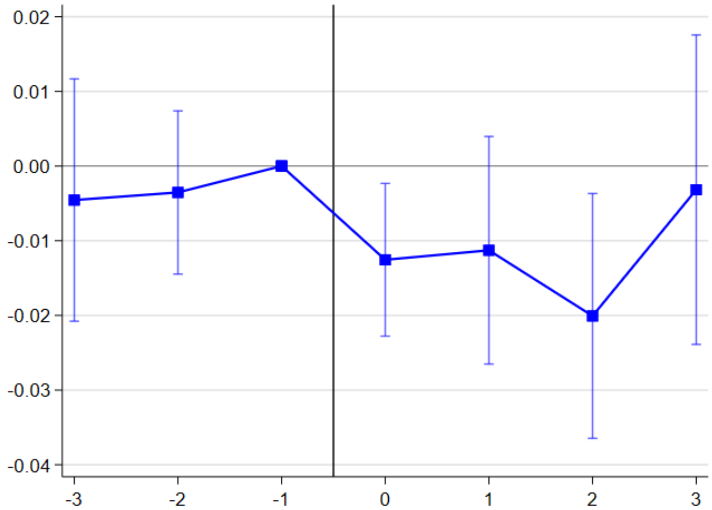
Figure 4: Distribution of Stringent Form



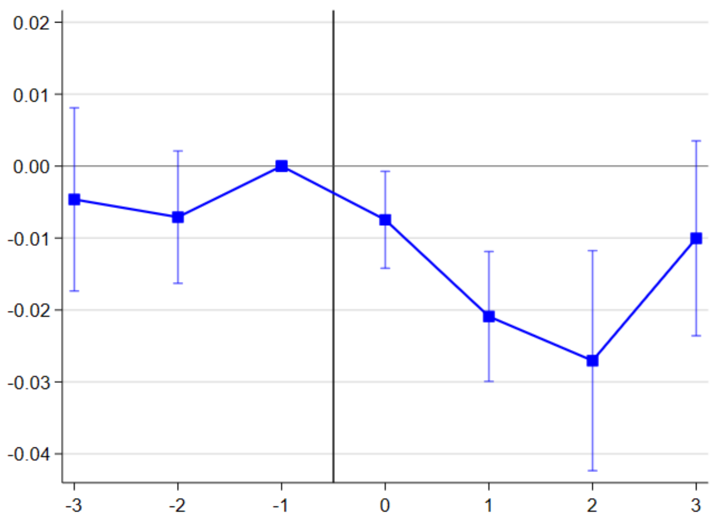
Note: The figure shows the distribution of *Stringent Form*, which is the proportion of premiums written under stringent form regulation for each firm-line observation.

Figure 5: Event Study Coefficients of Commercial Form Deregulation

A. Full Single-State Sample



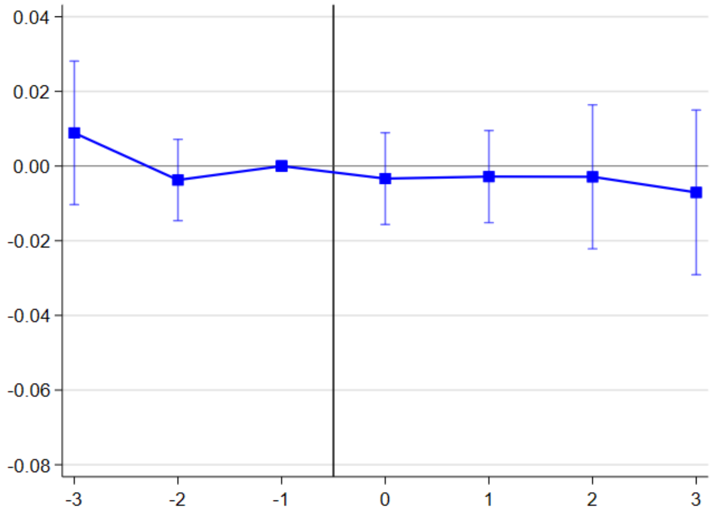
B. Matched Sample



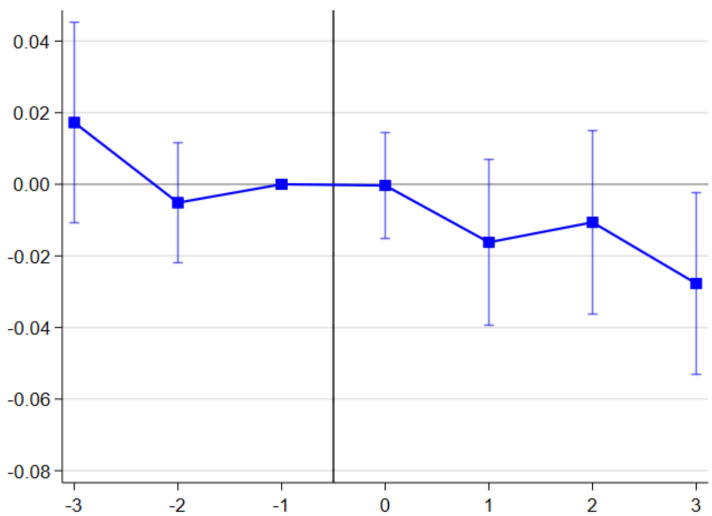
Notes: Panel A shows the estimated coefficients of the event-year indicators for the years since form de-regulation. We use single state firm-line-year observations writing in commercial-conventional lines. The dependent variable is the general expense ratio. Deregulation is an indicator variable that equals one for firm-lines operating in states that de-regulate commercial-conventional form (treatment group), and it equals zero for firm-lines always under prior approval form regulation (control group). The regression includes the interactions of the de-regulation indicator with event-year indicators, as well as event-year panel by firm-line fixed effects, and event-year panel by year fixed effects. The vertical bars represent the 95% confidence intervals. Panel B is estimated based on the matched sample. We identify a control group firm-line that were operating in the same rate regulatory environment as the treatment observations in the year prior to the de-regulation in each event panel.

Figure 6: Event Study Coefficients of Personal Form Deregulation

A. Full Single-State Sample

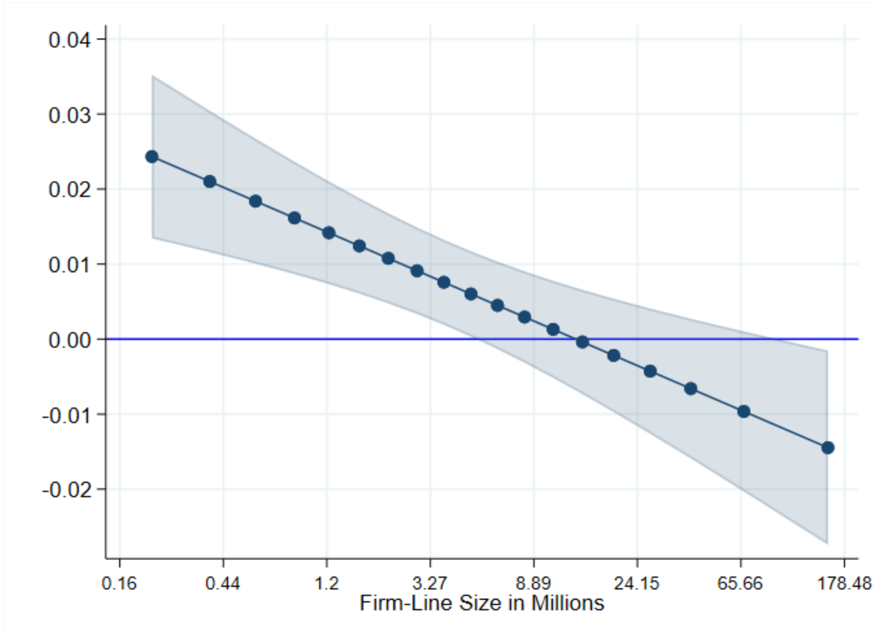


B. Matched Sample



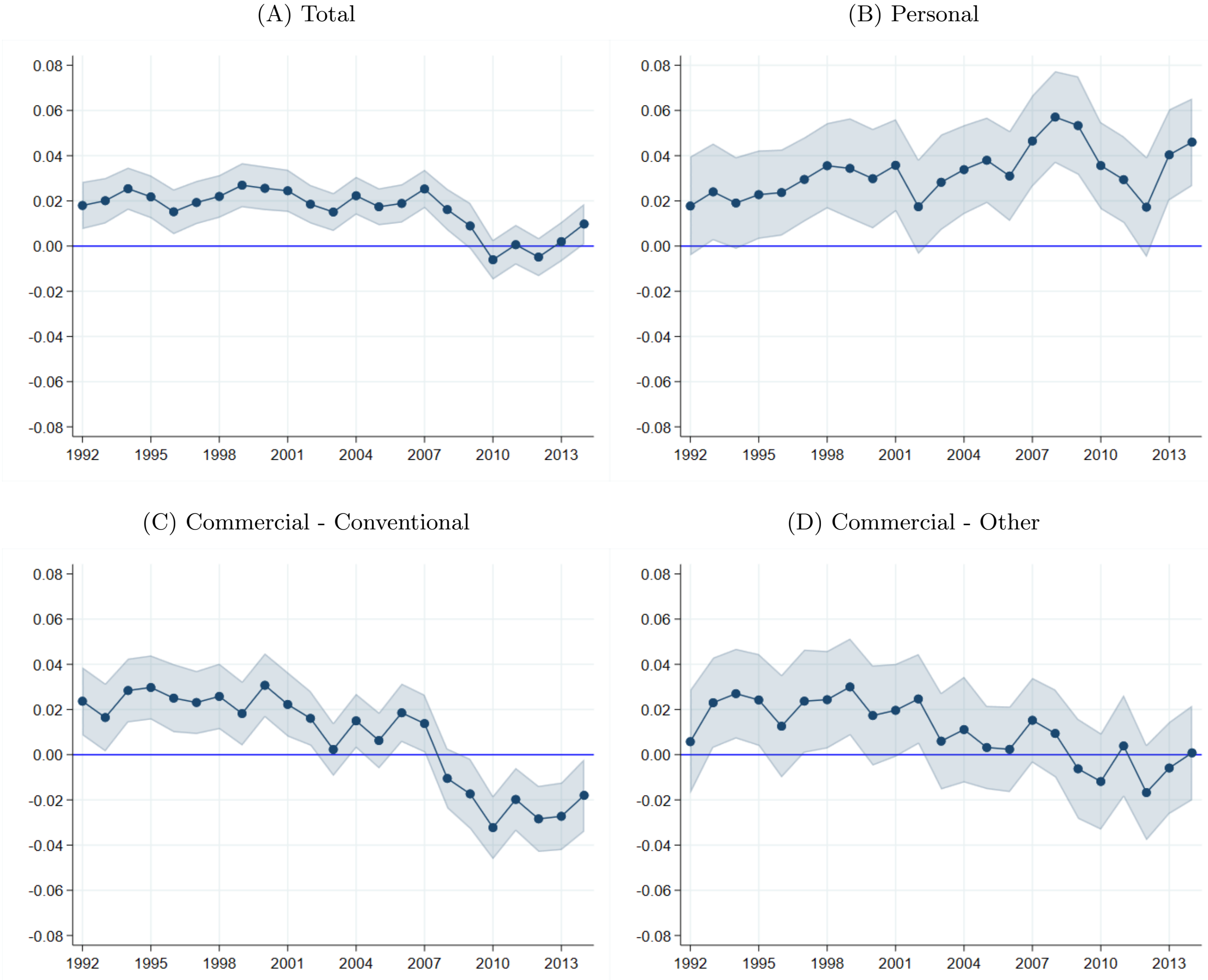
Notes: Panel A shows the estimated coefficients of the event-year indicators for the years since form de-regulation. We use single state firm-line-year observations writing in personal lines. The dependent variable is the general expense ratio. Deregulation is an indicator variable that equals one for firm-lines operating in states that de-regulate personal form (treatment group), and it equals zero for firm-lines always under prior approval form regulation (control group). The regression includes the interactions of the de-regulation indicator with event-year indicators, as well as event-year panel by firm-line fixed effects, and event-year panel by year fixed effects. The vertical bars represent the 95% confidence intervals. Panel B is estimated base on the matched sample. We identify a control group firm-line with the closest estimated propensity of the observed characteristics with a de-regulated firm-line in terms of the rate regulatory environment, total personal line premiums, and percent of commercial lines premiums written in a stringent form regulation. We use the data in the year prior to the de-regulation to calculate the propensity scores and match the control and the treatment in each event panel.

Figure 7: Marginal Effects of Stringent Form on General Expense Ratio by Firm-line Size (NPW)



Notes: The figure shows the estimated marginal effect of *Stringent Form* on the general expense ratio. The marginal effects are from estimating a regression model similar to Table 3, Column (2) with the inclusion of the interaction of *Stringent Form* and *Firm-line Size*. Firm-line size ranges from the 5th percentile to the 95th percentile, in increments of 5 percent (19 different points). Robust standard errors are clustered at the firm-line level. The shaded area represents the 95% confidence intervals.

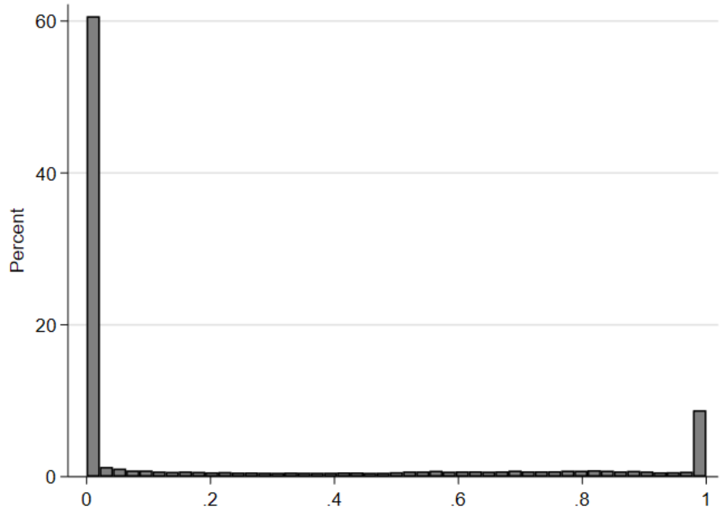
Figure 8: Marginal Effects of Stringent Form on the General Expense Ratio by Year and Line Type



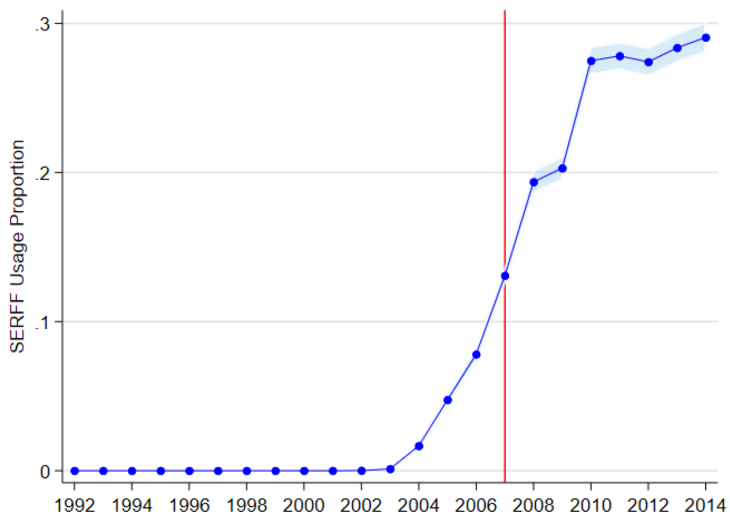
Notes: The figures show the estimated marginal effect of *Stringent Form* on the general expense ratio by year, separately for each line type. The marginal effects are estimated from a regression model similar to Table 3, Column (2); however, the model is expanded by including the interaction of *Stringent Form* with year indicators. Figure (A) is based on the full sample of firm-line-year observations. Figure (B) is based on the subsample of firm-lines that belong to the Personal line types. Figure (C) is based on the subsample of firm-lines that belong to the Commercial - Conventional line types. Figure (D) is based on the subsample of firm-lines that belong to the Commercial - Other line types. The shaded area represents the 95% confidence intervals. Robust standard errors are clustered at the firm-line level.

Figure 9: SERFF Usage

(A) Distribution

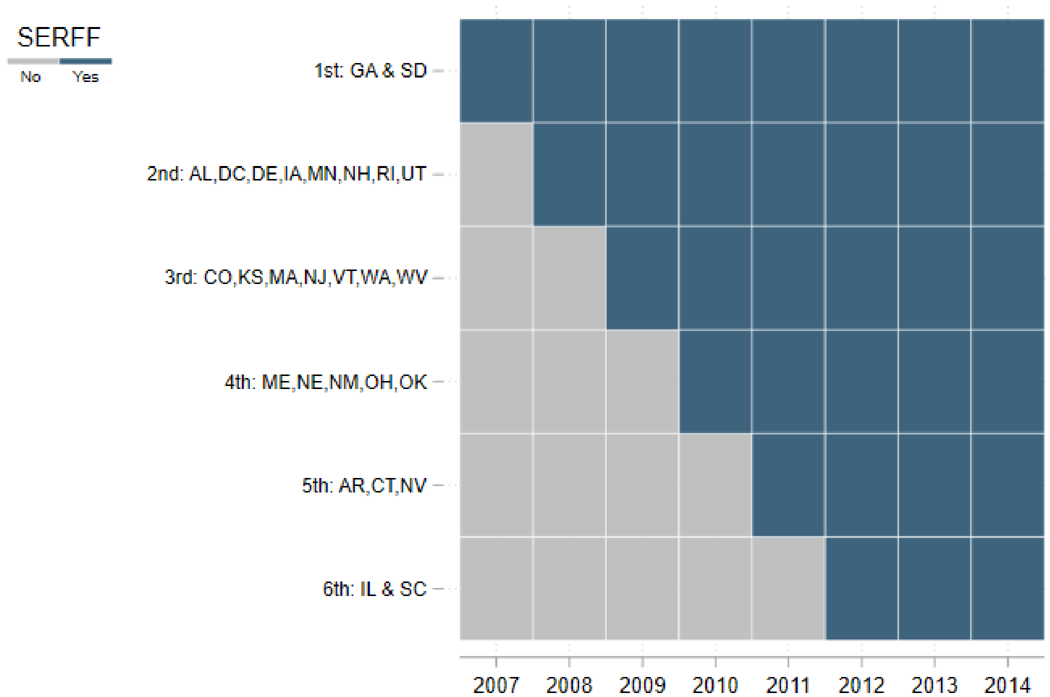


(B) Average by Year



Notes: Panel A shows the distribution of *SERFF Usage* for each firm-line-year observation. Panel B shows the average *SERFF Usage* across firm-lines, by year. SERFF usage is the proportion of premiums written under which the firm-line uses SERFF for form filings.
Data sources: Propriety data made available to the authors by the generosity of the NAIC.

Figure 10: SERFF Mandate by State



Note: The figure documents the year a state mandates the System for Electronic Rate and Form Filing (SERFF) for property-liability insurers.

Table 1: Summary Statistics

	Mean	SD	10^{th}	50^{th}	90^{th}
$Stringent\ Form_{i,l,t}$	0.583	0.394	0.000	0.687	1.000
$Stringent\ Rate_{i,l,t}$	0.263	0.348	0.000	0.077	0.988
$Number\ of\ States_{i,l,t}$	10.996	14.799	1.000	3.000	39.000
$Net\ Premiums\ Written_{i,l,t}$	44.672	331.655	0.388	4.837	67.515
$General\ Exp\ Ratio_{i,l,t}$	0.186	0.167	0.071	0.143	0.314
$Total\ Exp\ Ratio_{i,l,t}$	0.342	0.153	0.204	0.321	0.478
$Loss\ Ratio_{i,l,t}$	0.755	2.759	0.355	0.646	0.987
$Loss\ Volatility_{l,t}$	1.115	2.502	0.211	0.398	2.270
$Assets_{i,t}$	1,226.083	4,792.801	17.120	170.297	2,133.326
$State\ GDP\ Exposure_{i,l,t}$	473.096	363.821	123.789	381.538	923.045
$Population\ Exposure_{i,l,t}$	11.484	7.263	3.871	10.041	19.602
Observations	155,793				

Notes: The table shows the mean and standard deviation of the main variables at the firm(i)-line(l)-year(t) level (1992-2014). *Stringent Form (Rate)* is the proportion of premiums written under stringent form (rate regulation). *Number of States* is the number of states the firm-line is licensed to write business in the given year. *Net Premiums Written* is the total net premiums written in each firm-line in a given year in million dollars. *General Expense Ratio* is the ratio of general expenses to net premiums written. *Total Expense Ratio* is the ratio of all underwriting expenses (excluding loss adjustment expenses) to net premiums written. *Loss ratio* is the ratio of losses incurred to net premiums written. *Loss volatility* is the standard deviation of loss ratios of all the firms in a given line-year. *Assets* are total firm assets in million dollars. Data sources: NAIC (1992-2014) and state statutes. *State GDP Exposure* is the weighted average GDP (in billion dollars) across states for a firm-line-year, and *State Population Exposure* is the weighted average population across states for a firm-line-year (in millions).

Table 2: Summary Statistics by Line Types

	Mean	SD	10^{th}	50^{th}	90^{th}
Panel A: Personal Lines					
$Stringent\ Form_{i,l,t}$	0.772	0.340	0.004	0.979	1.000
$Stringent\ Rate_{i,l,t}$	0.359	0.389	0.000	0.202	1.000
$Number\ of\ States_{i,l,t}$	7.763	11.610	1.000	2.000	26.000
$Net\ Premiums\ Written_{i,l,t}$	85.278	579.137	0.809	8.916	113.502
$General\ Exp\ Ratio_{i,l,t}$	0.177	0.166	0.070	0.134	0.292
$Total\ Exp\ Ratio_{i,l,t}$	0.331	0.148	0.209	0.303	0.459
$Loss\ Ratio_{i,l,t}$	0.738	0.671	0.509	0.701	0.945
$Loss\ Volatility_{l,t}$	0.402	0.534	0.172	0.278	0.524
$State\ GDP\ Exposure_{i,l,t}$	453.751	385.927	105.526	347.052	958.706
$Population\ Exposure_{i,l,t}$	11.074	7.821	3.324	9.227	19.790
Observations	47,170				
Panel B: Commercial Lines - Conventional					
$Stringent\ Form_{i,l,t}$	0.482	0.383	0.000	0.485	1.000
$Stringent\ Rate_{i,l,t}$	0.169	0.278	0.000	0.012	0.598
$Number\ of\ States_{i,l,t}$	12.328	15.982	1.000	4.000	43.000
$Net\ Premiums\ Written_{i,l,t}$	24.566	97.240	0.322	3.347	45.154
$General\ Exp\ Ratio_{i,l,t}$	0.193	0.169	0.073	0.150	0.327
$Total\ Exp\ Ratio_{i,l,t}$	0.355	0.153	0.211	0.335	0.489
$Loss\ Ratio_{i,l,t}$	0.791	3.640	0.321	0.613	1.042
$Loss\ Volatility_{l,t}$	1.573	3.252	0.316	0.592	3.176
$State\ GDP\ Exposure_{i,l,t}$	481.625	352.156	131.715	396.944	913.444
$Population\ Exposure_{i,l,t}$	11.657	6.952	4.187	10.581	19.404
Observations	78,188				
Panel C: Commercial Lines - Other					
$Stringent\ Form_{i,l,t}$	0.549	0.393	0.000	0.655	1.000
$Stringent\ Rate_{i,l,t}$	0.357	0.379	0.000	0.213	1.000
$Number\ of\ States_{i,l,t}$	12.582	15.256	1.000	4.000	40.000
$Net\ Premiums\ Written_{i,l,t}$	33.394	123.390	0.315	4.469	74.407
$General\ Exp\ Ratio_{i,l,t}$	0.182	0.164	0.070	0.143	0.306
$Total\ Exp\ Ratio_{i,l,t}$	0.328	0.156	0.182	0.308	0.469
$Loss\ Ratio_{i,l,t}$	0.689	2.053	0.299	0.586	0.959
$Loss\ Volatility_{l,t}$	1.043	1.762	0.297	0.427	2.266
$State\ GDP\ Exposure_{i,l,t}$	481.168	356.808	135.569	391.691	916.941
$Population\ Exposure_{i,l,t}$	11.675	7.116	4.289	10.147	19.790
Observations	30,435				

Notes: The table shows the mean and standard deviation of the main variables at the firm(i)-line(l)-year(t) level (1992-2014), separately by line type. See Appendix Table A1 for the definitions of the line types. See Table 1 for definitions of the variables.

Table 3: The Effect of Stringent Form Regulation on the General Expense Ratio

	Baseline	Controls
	(1)	(2)
Stringent Form	0.0130***	0.0098***
	(0.0030)	(0.0030)
ln(Firm-Line Size)		-0.0732***
		(0.0017)
ln(Firm $Size_{i,t}$)		0.0133***
		(0.0028)
ln(No. States)		0.0246***
		(0.0017)
Stringent Rate		0.0058
		(0.0036)
Loss $Volatility_{l,t}$		-0.0001
		(0.0002)
ln(State GDP Exposure)		0.0150
		(0.0117)
ln(State Population Exposure)		-0.0067
		(0.0125)
Dep.Var. Mean	0.186	0.186
Stringent Form Mean	0.583	0.583
Year FE	Yes	Yes
Firm X Line FE	Yes	Yes
Within R^2	0.000	0.149
N	155,793	155,793

Notes: The table shows the results from estimating equation (1), which is a fixed effect regressions of the general expense ratio using firm-line-year observations (1992-2014). Column (1) reports the results of estimating equation (1) without time-varying control variables. Column (2) reports the results of estimating equation (1). *Stringent Form (Rate)* is the proportion of premiums written under stringent form (rate) regulation. *ln(Firm-Line Size)* is the natural logarithm of the net premiums written by an insurer in a line. *ln(Firm Size)* is the natural logarithm of the assets of an insurer. *ln(No. States)* is the natural logarithm of the number of states an insurer is licensed for a line. *Loss Volatility* is the standard deviation of loss ratios of all the firms in a given line-year. *ln(State GDP Exposure)* is logarithm values of the premium weighted average GDP across states for a firm-line-year, and *ln(State Population Exposure)* is logarithm values of the premium weighted average population across states for a firm-line-year. Robust standard errors, reported in parentheses, are clustered at the firm-line. *$p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Robustness Tests on General Expense Ratio

Panel A: Line by Year Fixed Effects		
	Baseline	Controls
	(1)	(2)
Stringent Form	0.0105***	0.0106***
	(0.0031)	(0.0031)
Dep.Var. Mean	0.186	0.186
Stringent Form Mean	0.583	0.583
Line X Year FE	Yes	Yes
Firm X Line FE	Yes	Yes
Controls	No	Yes
Within R^2	0.000	0.149
N	155,793	155,793

Panel B: Firm by Year Fixed Effects		
	(1)	(2)
Stringent Form	0.0105***	0.0131***
	(0.0025)	(0.0026)
Dep.Var. Mean	0.182	0.182
Stringent Form Mean	0.581	0.581
Firm X Year FE	Yes	Yes
Line FE	Yes	Yes
Controls	No	Yes
Within R^2	0.000	0.069
N	148,018	148,018

Notes: Number of observations in Panel B is smaller than the full sample (Table 3) due to the removal of singleton observations to avoid biasing standard errors (Correia, 2015). Robust standard errors, reported in parentheses, are clustered at the firm-line. *$p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 5: The Effect of De-regulating Forms in Commercial Lines

Panel A: Seven-year Window, -3 to +3 years			
	Baseline	Matched	
	(1)	(2)	(3)
Deregulate X Post	-0.0091	-0.0125***	-0.0125***
	(0.0064)	(0.0034)	(0.0034)
Dep.Var. Mean	0.158	0.152	0.152
Event Year X Firm X Line FE	Yes	Yes	No
Event Year X Firm FE	No	No	Yes
Event Year X Year FE	Yes	Yes	Yes
Within R^2	0.001	0.008	0.005
N	3,899	1,456	1,456

Panel B: Nine-year Window, -4 to +4 years (excludes 1995 deregulation)			
	Baseline	Matched	
	(1)	(2)	(3)
Deregulate X Post	-0.0182**	-0.0217***	-0.0217***
	(0.0068)	(0.0042)	(0.0042)
Dep.Var. Mean	0.153	0.147	0.147
Event Year X Firm X Line FE	Yes	Yes	No
Event Year X Firm FE	No	No	Yes
Event Year X Year FE	Yes	Yes	Yes
Within R^2	0.004	0.016	0.013
N	3,888	1,440	1,440

Notes: The table shows results from estimating a stacked differences-in-differences regression of the general expense ratio for commercial firm-lines that experience form de-regulation (prior approval to other form filing system) compared to firm-lines that are always under prior approval form filing system. The sample is comprised of single-state conventional commercial firm-lines. Panel A reports results using seven-year window and Panel B reports results using nine-year window (which excludes 1995 de-regulation event panel). Column (1) reports the results from estimating equation (2). Columns (2) and (3) report the results using matched sample. Column (3) replaces the event panel firm-line fixed effects used in Column (2) with event-panel firm fixed effects. Robust standard errors, reported in parentheses, are clustered at the state. *$p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 6: The Effect of De-regulating Forms in Personal Lines

Panel A: Seven-year Window, -3 to +3 years

	Baseline	Matched	
	(1)	(2)	(3)
Deregulate X Post	-0.0057	-0.0178*	-0.0178*
	(0.0058)	(0.0084)	(0.0084)
Dep.Var. Mean	0.185	0.168	0.168
Event Year X Firm X Line FE	Yes	Yes	No
Event Year X Firm FE	No	No	Yes
Event Year X Year FE	Yes	Yes	Yes
Within R^2	0.000	0.018	0.015
N	1,820	280	280

Panel B: Eight-year Window, -3 to +4 years

	Baseline	Matched	
	(1)	(2)	(3)
Deregulate X Post	-0.0043	-0.0153**	-0.0153**
	(0.0074)	(0.0066)	(0.0066)
Dep.Var. Mean	0.186	0.165	0.165
Event Year X Firm X Line FE	Yes	Yes	No
Event Year X Firm FE	No	No	Yes
Event Year X Year FE	Yes	Yes	Yes
Within R^2	0.000	0.011	0.010
N	1,912	320	320

Notes: The table shows results from estimating a stacked differences-in-differences regression of the general expense ratio for personal firm-lines that experience form de-regulation (prior approval to other form filing system) compared to firm-lines that are always under prior approval form filing system. The sample is comprised of single-state personal firm-lines. Panel A reports results using seven-year window and Panel B reports results using eight-year window. Column (1) reports the results from estimating equation (2). Columns (2) and (3) report the results using matched sample. Column (3) replaces the event panel firm-line fixed effects used in Column (2) with event-panel firm fixed effects. Robust standard errors, reported in parentheses, are clustered at the state. *$p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 7: The Effect of Stringent Form Regulation on the General Expense Ratio, by Lines

	PS	CM	CM-Other
	(1)	(2)	(3)
Stringent Form	0.0262***	0.0060*	0.0129*
	(0.0081)	(0.0036)	(0.0075)
ln(Firm-Line Size)	-0.0780***	-0.0723***	-0.0689***
	(0.0036)	(0.0023)	(0.0036)
ln(Firm $Size_{i,t}$*)*	0.0213***	0.0116***	0.0063
	(0.0057)	(0.0038)	(0.0056)
ln(No. States)	0.0234***	0.0257***	0.0224***
	(0.0033)	(0.0024)	(0.0037)
Stringent Rate	0.0063	0.0067	-0.0041
	(0.0060)	(0.0057)	(0.0080)
Loss $Volatility_{l,t}$	-0.0003	0.0000	-0.0010**
	(0.0012)	(0.0002)	(0.0005)
ln(State GDP Exposure)	-0.0123	0.0120	0.0670**
	(0.0218)	(0.0161)	(0.0263)
ln(State Population Exposure)	0.0226	-0.0085	-0.0513*
	(0.0237)	(0.0171)	(0.0275)
Dep.Var. Mean	0.177	0.193	0.182
Stringent Form Mean	0.772	0.482	0.549
Year FE	Yes	Yes	Yes
Firm X Line FE	Yes	Yes	Yes
Within R^2	0.171	0.140	0.142
N	47,170	78,188	30,435

Notes: The table shows the results from estimating equation (1), which is a fixed effect regressions of the general expense ratio using firm-line-year observations (1992-2014), by lines. Column (1) uses the subsample of personal (PS) lines. Column (2) uses the subsample of conventional commercial lines (CM). Column (3) uses the subsample of other commercial lines ($CM - Other$). For definitions of the variables, see Table 3. Robust standard errors, reported in parentheses, are clustered at the firm-line. *$p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Days till Disposition and Stringent Form Regulation

	Baseline		PS	CM	CM-Other
	(1)	(2)	(3)	(4)	(5)
I(Stringent Form$_{l,s,t}$)	4.5277	7.7133*	6.0693	9.8210**	0.2489
	(4.5182)	(3.8653)	(4.6918)	(4.7907)	(3.0738)
ln(Firm-line-state Size$_{i,l,s,t}$)		0.5758**	1.1278***	0.4706	0.1987
		(0.2348)	(0.3105)	(0.3066)	(0.2255)
ln(Firm-line Size$_{i,l,t}$)		-0.6438**	-1.0459*	-0.4372	-0.4882
		(0.2586)	(0.5424)	(0.3288)	(0.5220)
ln(Firm Size$_{i,t}$)		-0.2118	-1.8576	0.2126	-0.5528
		(0.5347)	(1.4750)	(0.5989)	(1.0206)
I(Stringent Rate$_{l,s,t}$)		-6.8138	-13.3408**	-5.7163	-2.6811
		(5.5245)	(5.3596)	(10.5893)	(3.1957)
ln(All filings$_{l,s,t}$)		-4.8493	-12.6123**	-5.4349	-0.7646
		(3.0357)	(5.2584)	(3.8089)	(2.5427)
ln(State GDP$_{s,t}$)		29.6672**	30.9450**	31.3549***	19.5089*
		(11.3888)	(13.6345)	(11.6840)	(10.0976)
ln(State Population$_{s,t}$)		-22.2673*	-20.9718	-23.4572*	-13.7449
		(11.7946)	(14.3487)	(12.0949)	(10.3521)
Dep.Var. Mean	25.946	25.946	29.582	25.293	24.747
Share Stringent Form	0.446	0.446	0.714	0.351	0.507
Firm X Line FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Within R^2	0.004	0.073	0.071	0.093	0.040
N	252,547	252,547	44,778	158,066	49,703

Notes: The table shows the results from estimating fixed effect regressions of the days till disposition using firm-line-state-year observations for form filings. The sample includes firm-line-state-year observations from 2007 to 2014. *ln(Firm-line-state Size$_{i,l,s,t}$)* is logarithm values of total direct premiums written in a firm-line-state-year, *ln(All filings$_{l,s,t}$)* is logarithm values of total number of filings (including all types of filings) submitted by insurers (including unlicensed and out-of-sample insurers) for a given line-state-year, *ln(State GDP$_{s,t}$)* is logarithm values of total GDP in a state-year, and *ln(State Population$_{s,t}$)* is logarithm values of total number of population in a state-year. Robust standard errors, reported in parentheses, are clustered at the state. *$p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Days till Disposition and Stringent Form Regulation, Firm X Line X State FE

	Baseline		PS	CM	CM-Other
	(1)	(2)	(3)	(4)	(5)
I(Stringent Form$_{l,s,t}$)	18.0979	16.8510*	5.8613***	17.0531*	20.4206
	(11.0519)	(9.9558)	(1.8354)	(9.9139)	(12.7839)
ln(Firm-line-state Size$_{i,l,s,t}$)		-0.4925***	-1.5038***	-0.2372	-0.4158
		(0.1597)	(0.5028)	(0.1541)	(0.3420)
ln(Firm-line Size$_{i,l,t}$)		-0.0304	0.6471	-0.1214	-0.6261
		(0.2065)	(0.5198)	(0.2710)	(0.5074)
ln(Firm Size$_{i,t}$)		-0.2268	-1.2937	0.3073	-0.2000
		(0.5589)	(1.3367)	(0.6482)	(1.2851)
I(Stringent Rate$_{l,s,t}$)		7.3449	6.3166	7.6284	17.9687***
		(7.2941)	(6.3957)	(9.0308)	(2.8597)
ln(All filings$_{l,s,t}$)		-10.0945***	-8.1919	-13.4080***	-5.3347**
		(3.3692)	(5.6584)	(4.5636)	(2.4829)
ln(State GDP$_{s,t}$)		33.6775	26.4874	36.2607	27.0586
		(34.0741)	(33.8032)	(37.1947)	(32.9481)
ln(State Population$_{s,t}$)		52.6853	48.3234	78.0027	-18.8019
		(92.6179)	(95.5272)	(103.1577)	(92.9099)
Dep.Var. Mean	25.939	25.939	29.475	25.264	24.834
Share Stringent Form	0.447	0.447	0.713	0.352	0.508
Firm X Line X State FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Within R^2	0.004	0.011	0.004	0.017	0.006
N	239,780	239,780	43,103	151,150	45,527

Notes: The table shows the results from estimating fixed effect regressions similar to those in Table 8, except we include firm-by-line-by-state fixed effects. Robust standard errors, reported in parentheses, are clustered at the state. $*p < 0.1$, $**\ p < 0.05$, $***\ p < 0.01$

Table 10: The Effect of SERFF Usage on the General Expense Ratio

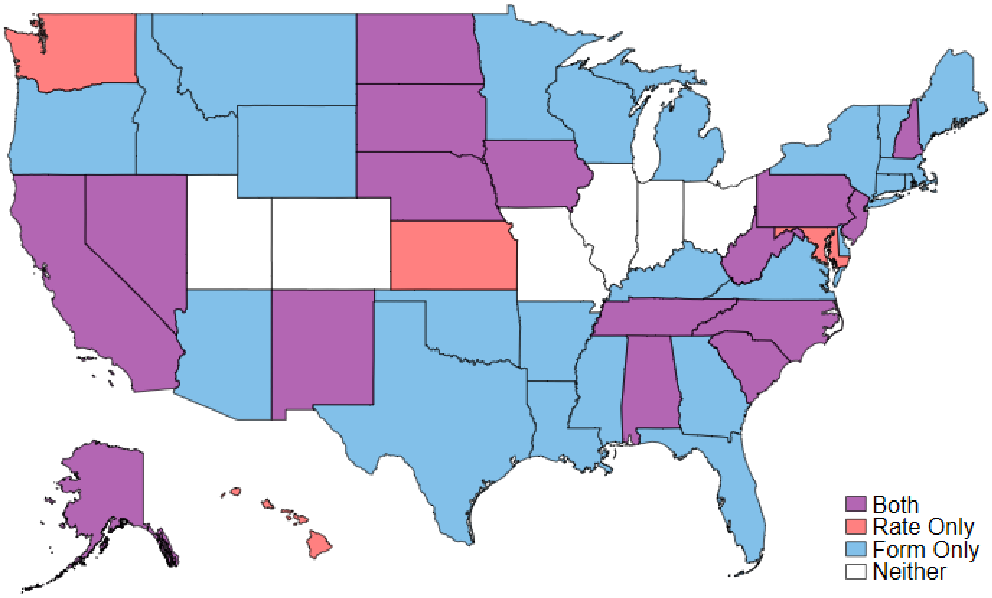
	Baseline		Firm SE Cluster	
	(1)	(2)	(3)	(4)
I(SERFF Usage)	-0.0367***	-0.0433***	-0.0367**	-0.0433***
	(0.0101)	(0.0113)	(0.0152)	(0.0166)
Stringent Form	0.0050	0.0018	0.0050	0.0018
	(0.0034)	(0.0040)	(0.0045)	(0.0051)
Stringent Form X I(SERFF)		0.0143		0.0143
		(0.0097)		(0.0125)
Stringent Rate	0.0038	0.0044	0.0038	0.0044
	(0.0049)	(0.0049)	(0.0065)	(0.0064)
ln(No. States)	0.0243***	0.0243***	0.0243***	0.0243***
	(0.0028)	(0.0028)	(0.0040)	(0.0040)
ln(Firm-Line Size)	-0.0806***	-0.0806***	-0.0806***	-0.0806***
	(0.0025)	(0.0025)	(0.0035)	(0.0035)
Loss $Volatility_{l,t}$	-0.0001	-0.0002	-0.0001	-0.0002
	(0.0002)	(0.0002)	(0.0002)	(0.0002)
ln(Firm $Size_{i,t}$*)*	0.0136***	0.0137***	0.0136**	0.0137**
	(0.0041)	(0.0041)	(0.0067)	(0.0068)
ln(State GDP Exposure)	0.0075	0.0074	0.0075	0.0074
	(0.0145)	(0.0145)	(0.0187)	(0.0188)
ln(State Population Exposure)	0.0034	0.0034	0.0034	0.0034
	(0.0156)	(0.0156)	(0.0197)	(0.0197)
Dep. Var. Mean	0.188	0.188	0.188	0.188
Full Effects of Stringent Form		0.016*		0.016
First-stage KP F-stat	956.431	475.986	492.562	249.799
Year FE	Yes	Yes	Yes	Yes
Firm X Line FE	Yes	Yes	Yes	Yes
Within R^2	0.141	0.141	0.141	0.141
N	82,936	82,936	82,936	82,936

Notes: The table shows second-stage results from estimating the two-stage instrumental variable regression model outlined in equations (4)-(5). First-stage results are reported in Appendix Table A13. The sample includes firm-line-year observations from 2002 to 2014. Column (2) adds the interaction of *Stringent Form* and *I(SERFF Usage)* to the model in column (1). *I(SERFF Usage)* is an indicator variable that equals to one if the firm-line used SERFF in a given year and zero otherwise. See Table 3 for definitions of other variables. Robust standard errors, reported in parentheses, are clustered at the firm-line level. *$p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

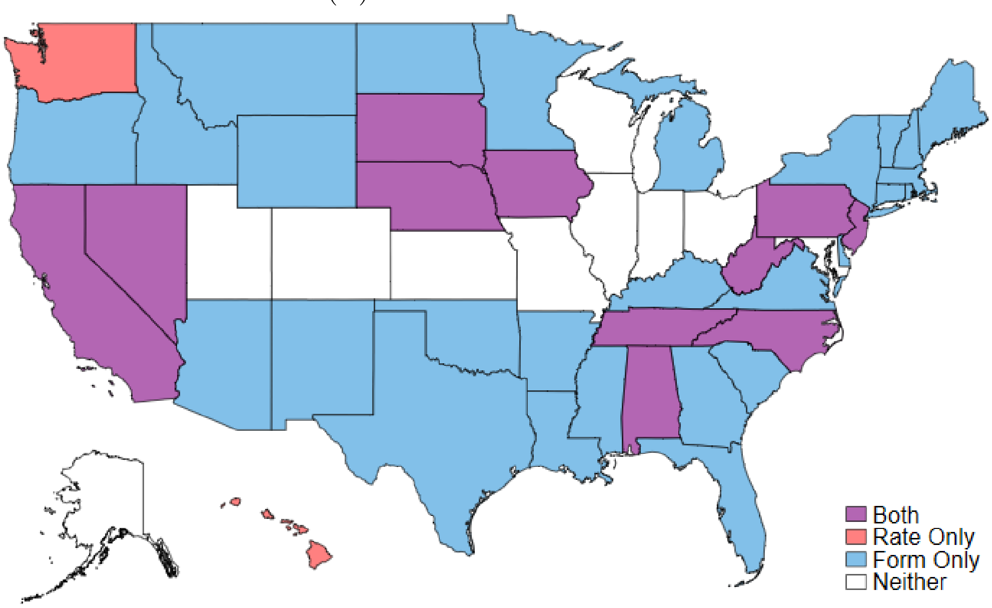
Appendix

Figure A1: Stringent Form X Rate Regulation, Personal

(A) Personal Lines in 1992

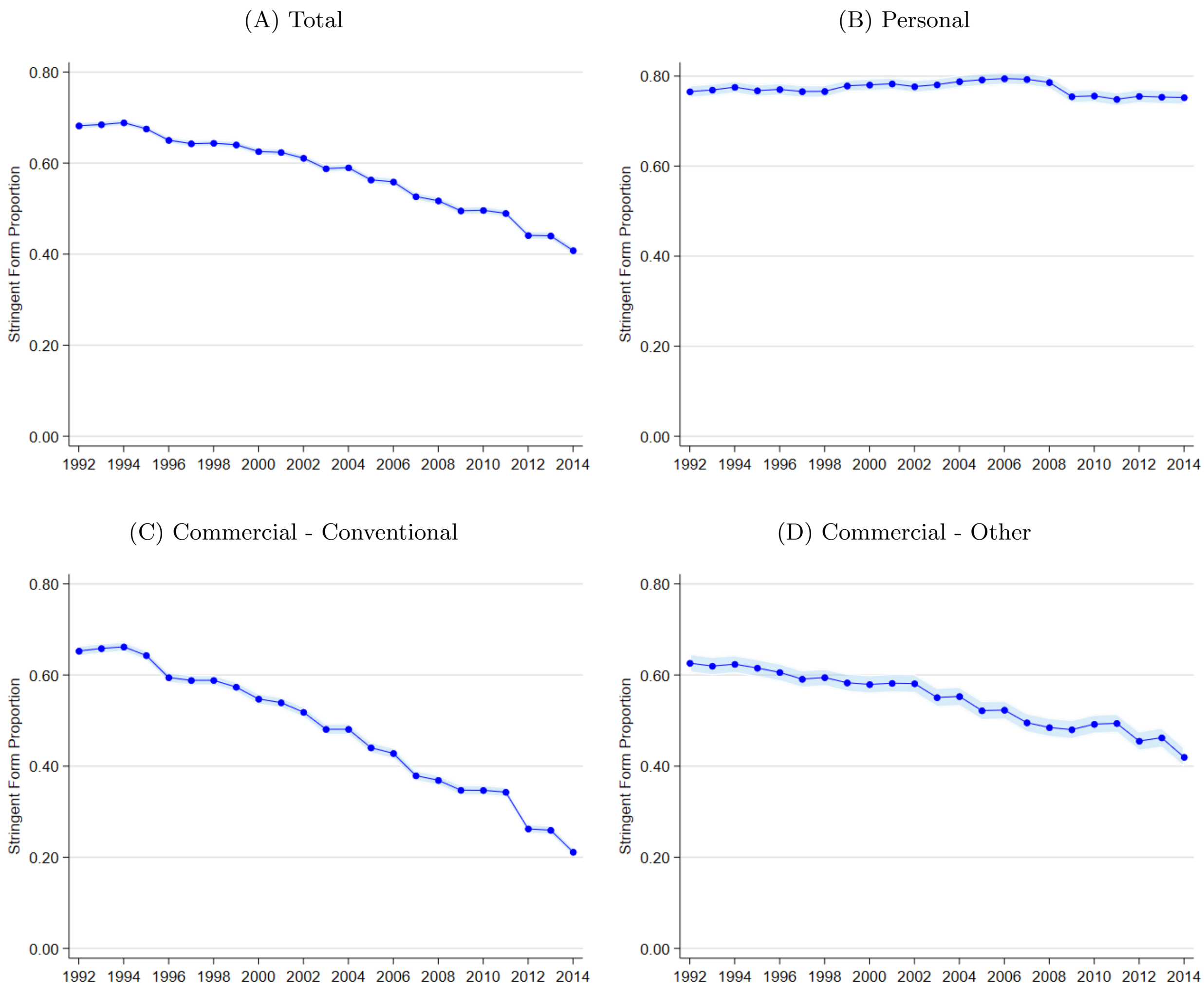


(B) Personal Lines in 2014



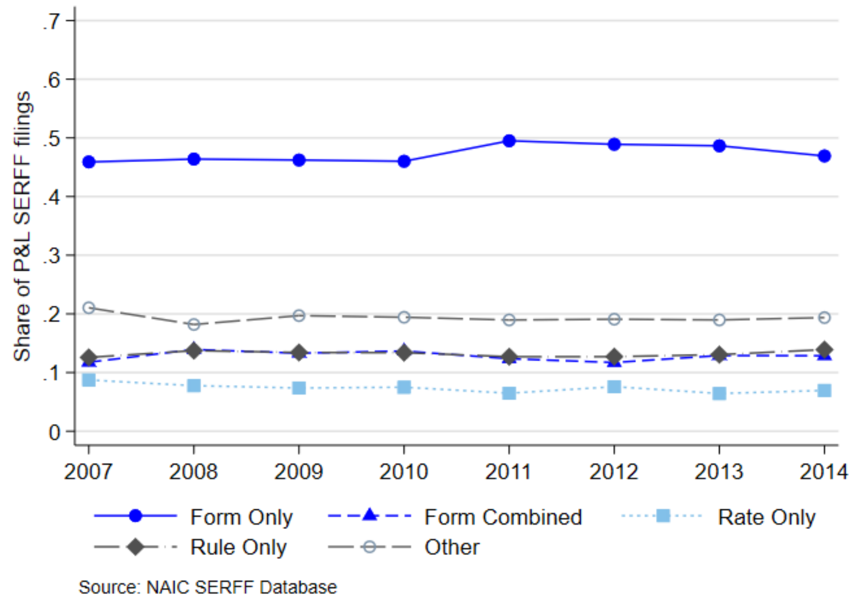
Notes: The figures show whether forms and/or rates of the personal lines are regulated under "Prior Approval" in 1992 (figure (a)) and 2014 (figure (b) . Data sources: NAIC (1992, 2014) and state statutes.

Figure A2: Stringent Form by Line and Year



Notes: The figures show the average *Sringent Form* by year. Figure (A) shows the full sample of firm-line-year observations, Figure (B) the Personal lines, Figure (C) the Commercial - Conventional lines, and Figure (D) the Commercial - Other lines. The shaded area represents the 95% confidence intervals.

Figure A3: SERFF Form and Rate Filings



Note: The figure shows the share of form, rate, and other filings with the SERFF system. *Form Only* is the share of SERFF filings that includes form filings but no rate or rule filings. *Rate Only* is the share of SERFF filings that includes rate filings but no form or rule filings. *Form & Rate* is the share of SERFF filings that includes both rate and form filings (filings may also include rule submissions). *Other* is the share of SERFF filings pertaining to non-rate and non-form filings, e.g., filings for certification and licensing, rule only filings. Data sources: NAIC SERFF Database.

Table A1: Line of Insurance Category

Line Type	Line of Insurance in the Sample	Original Line in NAIC Data	Note
Personal	Homeowners/ Farmowners	Farmowners multiple peril	
		Homeowners multiple peril	
	Private Passenger Auto Liability	Private passenger auto no-fault (personal injury protection)	
		Other private passenger auto liability	
	Private Passenger Auto Physical Damage	Privatepassenger auto physical damage	
Commercial - conventional	Special Property	Fire	
		Allied lines	
		Earthquake	
		Glass	
		Burglary and theft	
	Commercial Multiple Peril	Commercial multiple peril (non-liability portion)	
		Commercial multiple peril (liability portion)	
	Financial /Mortgage Guaranty	Mortgage guaranty	Not used
		Financial guaranty	Not used
	Other Liability	Other liability	
		Other liability - occurrence	
		Other liability - claims made	
	Products Liability	Products Liability	
	Commercial Auto Liability	Commercial auto no-fault (personal injury protection)	
		Other commercial auto liability	
	Commercial Auto Physical Damage	Commercial auto physical damage	
	Fidelity/Surety	Fidelity	Not used
		Surety	Not used
	Special Liability	Aircraft (all perils)	
		Boiler and machinery	
	Credit	Credit	Not used
	Warranty	Warranty	Not used
Commercial - other	Workers' compensation	Workers' compensation	
	Medical Professional Liability	Medical Professional Liability	
	Ocean Marine	Ocean Marine	
	Inland Marine	Inland Marine	

Notes: The table shows the categorization of the lines of business.

[For Online Publication]

Table A2: Description of the Major Form Filing Systems

Type	Description
Prior Approval	Forms must be filed with and approved by the state regulator before they can be used. There may be a "deemer" policy, which means forms are considered approved if not denied within a certain number of days.
File and Use	Forms must be filed with the state regulator a certain number of days prior to their use. Approval is not required.
Use and File	Forms must be filed with the state regulator within a certain number of days after they have been used.
File Only	Forms need to be filed but the deadline of the filing is not specified by statute.
No Filing	Forms are not required to be filed.

Notes: The table shows the description of the major form filing systems in the U.S. property and liability insurance market. Data sources: NAIC (1992-2014) and state statutes.

Table A3: Form Regulation in Commercial Lines by State, 1992-2014

State	Stringent Form Regulation	Non-stringent Form Regulation	State	Stringent Form Regulation	Non-stringent Form Regulation
Alabama	1992-2001	2002-2014	Montana	1992-2014	
Alaska	1992-2004	2005-2014	Nebraska		1992-2014
Arizona	1992-1998	1999-2014	Nevada	1992-2014	
Arkansas	1992-1999	2000-2014	New Hampshire	1992-1998	1999-2014
California	1992-2014		New Jersey		1992-2014
Colorado		1992-2014	New Mexico	1992-2005	2006-2014
Connecticut	1992-2014		New York	1992-2011	2012-2014
Delaware	1992-2014		North Carolina	1992-2014	
District of Columbia		1992-2014	North Dakota	1992-2014	
Florida	1992-2013	2014	Ohio		1992-2014
Georgia	1992-2014		Oklahoma	1992-2014	
Hawaii		1992-2014	Oregon	1992-2014	
Idaho	1992-1994	1995-2014	Pennsylvania	1992-1998	1999-2014
Illinois		1992-2014	Rhode Island[1]	1998	1999-2014
Indiana		1992-2014	South Carolina	1992-2002	2003-2014
Iowa	1992-2014		South Dakota	1992-2004	2005-2014
Kansas		1992-2014	Tennessee		1992-2014
Kentucky	1992-2014		Texas	1992-2006	2007-2014
Louisiana	1992-1999	2000-2014	Utah		1992-2014
Maine	1992-1999	2000-2014	Vermont	1992-2014	
Maryland		1992-2014	Virginia	1992-2000	2001-2014
Massachusetts	1992-2004	2005-2014	Washington		1992-2014
Michigan	1992-2002	2003-2014	West Virginia	1992-2005	2006-2014
Minnesota	1992-1994	1995-2014	Wisconsin	1992-2008	2009-2014
Mississippi	1992-2014		Wyoming	1992-2014	
Missouri		1992-2014			

1. Data missing during 1992-1997.

Notes: The table shows how policy forms are regulated in conventional commercial insurance lines by state for the years 1992-2014. Conventional commercial lines are: special property, commercial multiple peril, other liability, products liability, commercial auto liability, commercial auto physical damage, and special liability. Stringent form regulation is identified by a prior approval form filing system. Data sources: NAIC (1992-2014) and state statutes.

Table A4: Form Regulation in Personal Lines by State, 1992-2014

State	Stringent Form Regulation	Non-stringent Form Regulation	State	Stringent Form Regulation	Non-stringent Form Regulation
Alabama	1992-2014		Montana	1992-2014	
Alaska	1992-2005	2006-2014	Nebraska	1992-2014	
Arizona	1992-2014		Nevada	1992-2014	
Arkansas	1992-2014		New Hampshire	1992-2014	
California	1992-2014		New Jersey	1992-2014	
Colorado		1992-2014	New Mexico	1992-2014	
Connecticut	1992-2014		New York	1992-2014	
Delaware	1992-2014		North Carolina	1992-2014	
District of Columbia		1992-2014	North Dakota	1992-2014	
Florida	1992-2014		Ohio		1992-2014
Georgia	1992-2014		Oklahoma	1992-2014	
Hawaii		1992-2014	Oregon	1992-2014	
Idaho	1992-1994	1995-2014	Pennsylvania	1992-2014	
Illinois		1992-2014	Rhode Island[1]	1998-2014	
Indiana		1992-2014	South Carolina	1992-2014	
Iowa	1992-2014		South Dakota	1992-2014	
Kansas		1992-2014	Tennessee	1992-2014	
Kentucky	1992-2014		Texas	1992-2014	
Louisiana	1992-2014		Utah		1992-2014
Maine	1992-2014		Vermont	1992-2014	
Maryland		1992-2014	Virginia	1992-2014	
Massachusetts	1992-2014		Washington		1992-2014
Michigan	1992-2014		West Virginia	1992-2014	
Minnesota	1992-2014		Wisconsin	1992-2008	2009-2014
Mississippi	1992-2014		Wyoming	1992-2014	
Missouri		1992-2014			

1. Data missing during 1992-1997.

Notes: The table shows how policy forms are regulated in personal insurance lines by state for the years 1992-2014. Personal lines are: homeowners/farmowners, private passenger auto liability, and private passenger auto physical damage. Stringent form regulation is identified by a prior approval form filing system. Data sources: NAIC (1992-2014) and state statutes.

Table A5: Rate Regulation in Commercial Lines by State, 1992-2014

State	Stringent Rate Regulation	Non-stringent Rate Regulation	State	Stringent Rate Regulation	Non-stringent Rate Regulation
Alabama	1992-2001	2002-2014	Montana		1992-2014
Alaska	1992-2005	2006-2014	Nebraska		1992-2014
Arizona		1992-2014	Nevada	1992-1993	1994-2014
Arkansas[1]		1994-2014	New Hampshire		1992-2014
California	1992-2014		New Jersey		1992-2014
Colorado		1992-2014	New Mexico	1992-2007	2008-2014
Connecticut		1992-2014	New York		1992-2014
Delaware		1992-2014	North Carolina		1992-2014
District of Columbia	1991-2000	2001-2014	North Dakota	1992-2007	2008-2014
Florida		1992-2014	Ohio		1992-2014
Georgia		1992-2014	Oklahoma	1992-1999	2000-2014
Hawaii	1992-2014		Oregon		1992-2014
Idaho		1992-2014	Pennsylvania	1992-1998	1999-2014
Illinois		1992-2014	Rhode Island		1992-2014
Indiana		1992-2014	South Carolina[3]	1995-1999	2000-2014
Iowa	1992-2014		South Dakota	1992-2004	2005-2014
Kansas[2]	1997-1999	2000-2014	Tennessee		1992-2014
Kentucky		1992-2014	Texas		1992-2014
Louisiana		1992-2014	Utah		1992-2014
Maine		1992-2014	Vermont		1992-2014
Maryland	1992-1997	1998-2014	Virginia		1992-2014
Massachusetts		1992-2014	Washington	1992-1997	1998-2014
Michigan		1992-2014	West Virginia	1992-2005	2006-2014
Minnesota		1992-2014	Wisconsin		1992-2014
Mississippi		1992-2014	Wyoming		1992-2014
Missouri		1992-2014			

1. Data missing during 1992-1993.
2. Data missing during 1992-1996.
3. Data missing during 1992-1994.

Notes: The table shows how rates are regulated in conventional commercial insurance lines by state for the years 1992 to 2014. Conventional commercial lines are: special property, commercial multiple peril, other liability, products liability, commercial auto liability, commercial auto physical damage, and special liability. Stringent rate regulation is identified by a prior approval rate filing system. In a few states, commercial auto insurance is sometimes regulated differently than other commercial lines. While it is not reflected in the Table, for parsimony, rate regulation for commercial auto insurance is appropriately coded in the data. Data sources: NAIC (1992-2014) and state statutes.

Table A6: Rate Regulation in Personal Lines by State, 1992-2014

State	Stringent Rate Regulation	Non-stringent Rate Regulation	State	Stringent Rate Regulation	Non-stringent Rate Regulation
Alabama	1992-2014		Montana		1992-2014
Alaska	1992-2005	2006-2014	Nebraska	1992-2014	
Arizona		1992-2014	Nevada	1992-2014	
Arkansas[1]		1994-2014	New Hampshire	1992-2003	2004-2014
California	1992-2014		New Jersey	1992-2014	
Colorado		1992-2014	New Mexico	1992-2007	2008-2014
Connecticut		1992-2014	New York		1992-2014
Delaware		1992-2014	North Carolina	1992-2014	
District of Columbia		1992-2014	North Dakota	1992-2007	2008-2014
Florida		1992-2014	Ohio		1992-2014
Georgia		1992-2014	Oklahoma		1992-2014
Hawaii	1992-2014		Oregon		1992-2014
Idaho		1992-2014	Pennsylvania	1992-2014	
Illinois		1992-2014	Rhode Island		1992-2014
Indiana		1992-2014	South Carolina[3]	1995-2004	2005-2014
Iowa	1992-2014		South Dakota	1992-2004	2005-2014
Kansas[2]	1997-1999	2000-2014	Tennessee	1992-2014	
Kentucky		1992-2014	Texas		1992-2014
Louisiana	1995-2007	2008-2014	Utah		1992-2014
Maine		1992-2014	Vermont		1992-2014
Maryland	1992-1997	1998-2014	Virginia		1992-2014
Massachusetts		1992-2014	Washington	1992-2014	
Michigan		1992-2014	West Virginia	1992-2014	
Minnesota		1992-2014	Wisconsin		1992-2014
Mississippi		1992-2014	Wyoming		1992-2014
Missouri		1992-2014			

1. Data missing during 1992-1993.
2. Data missing during 1992-1996.
3. Data missing during 1992-1994.

Notes: The table shows how rates are regulated in personal insurance lines by state for the years 1992-2014. Personal lines are: homeowners/farmowners, private passenger auto liability, and private passenger auto physical damage. Stringent rate regulation is identified by a prior approval rate filing system. In a few states, personal auto insurance is sometimes regulated differently than other personal lines. While it is not reflected in the Table, for parsimony, rate regulation for personal auto insurance is appropriately coded in the data. Data sources: NAIC (1992-2014) and state statutes.

Table A7: The Effect of Stringent Form Regulation on *ln(General Expenses)*

	Main		PS	CM	CM-Other
	(1)	(2)	(3)	(4)	(5)
Stringent Form	0.0731***	0.0594***	0.0864**	0.0429**	0.0811**
	(0.0248)	(0.0142)	(0.0384)	(0.0181)	(0.0348)
ln(Firm-Line Size)		0.7101***	0.7217***	0.7063***	0.7091***
		(0.0069)	(0.0128)	(0.0092)	(0.0165)
ln(Firm $Size_{i,t}$*)*		0.0479***	0.0095	0.0686***	0.0524**
		(0.0119)	(0.0221)	(0.0171)	(0.0253)
ln(No. States)		0.1299***	0.1317***	0.1331***	0.1200***
		(0.0086)	(0.0174)	(0.0116)	(0.0187)
Stringent Rate		0.0345**	0.0235	0.0319	0.0118
		(0.0168)	(0.0266)	(0.0266)	(0.0368)
Loss $Volatility_{l,t}$		-0.0013*	-0.0013	-0.0010	-0.0019
		(0.0007)	(0.0050)	(0.0008)	(0.0022)
ln(State GDP Exposure)		0.0960	-0.1118	0.1101	0.3961***
		(0.0585)	(0.1018)	(0.0853)	(0.1297)
ln(State Population Exposure)		-0.0570	0.1709	-0.0902	-0.3290**
		(0.0625)	(0.1126)	(0.0900)	(0.1360)
Dep.Var. Mean	13.520	13.520	14.069	13.227	13.423
Stringent Form Mean	0.583	0.583	0.772	0.482	0.549
Year FE	Yes	Yes	Yes	Yes	Yes
Firm X Line FE	Yes	Yes	Yes	Yes	Yes
Within R^2	0.000	0.521	0.542	0.509	0.519
N	155,793	155,793	47,170	78,188	30,435

Notes: The table shows the results from estimating models in Table 3 with an outcome variable equal to the natural logarithm of general expenses (in dollar). Robust standard errors, reported in parentheses, are clustered at the firm-line level. $*p < 0.1$, $**\ p < 0.05$, $***\ p < 0.01$

Table A8: The Effect of Stringent Form Regulation on the General Expense Ratio, Firm Clusters

	Main		PS	CM	CM-Other
	(1)	(2)	(3)	(4)	(5)
Stringent Form	0.0130***	0.0098**	0.0262**	0.0060	0.0129*
	(0.0041)	(0.0041)	(0.0105)	(0.0051)	(0.0076)
ln(Firm-Line Size)		-0.0732***	-0.0780***	-0.0723***	-0.0689***
		(0.0024)	(0.0046)	(0.0029)	(0.0037)
ln(Firm $Size_{i,t}$*)*		0.0133***	0.0213***	0.0116**	0.0063
		(0.0046)	(0.0076)	(0.0059)	(0.0061)
ln(No. States)		0.0246***	0.0234***	0.0257***	0.0224***
		(0.0025)	(0.0044)	(0.0033)	(0.0038)
Stringent Rate		0.0058	0.0063	0.0067	-0.0041
		(0.0048)	(0.0079)	(0.0071)	(0.0081)
Loss $Volatility_{l,t}$		-0.0001	-0.0003	0.0000	-0.0010**
		(0.0002)	(0.0013)	(0.0002)	(0.0005)
ln(State GDP Exposure)		0.0150	-0.0123	0.0120	0.0670***
		(0.0155)	(0.0272)	(0.0196)	(0.0254)
ln(State Population Exposure)		-0.0067	0.0226	-0.0085	-0.0513*
		(0.0162)	(0.0296)	(0.0203)	(0.0267)
Dep.Var. Mean	0.186	0.186	0.177	0.193	0.182
Stringent Form Mean	0.583	0.583	0.772	0.482	0.549
Year FE	Yes	Yes	Yes	Yes	Yes
Firm X Line FE	Yes	Yes	Yes	Yes	Yes
Within R^2	0.000	0.149	0.171	0.140	0.142
N	155,793	155,793	47,170	78,188	30,435

Notes: The table shows the results from estimating models in Table 3 clustering standard errors at the firm level. $*p < 0.1$, $**\ p < 0.05$, $***\ p < 0.01$

Table A9: Descriptive Statistics of the Stacked Panels - Commercial Lines

Panel A: Summary Statistics

	Mean	SD	10^{th}	50^{th}	90^{th}
General Exp Ratio	0.158	0.094	0.071	0.134	0.267
Stringent Form	0.886	0.317	0.000	1.000	1.000
I(Stringent Rate)	0.301	0.459	0.000	0.000	1.000
Net Premiums Written	5.729	12.215	0.433	2.466	12.996
$Loss\ Volatility_{l,t}$	1.560	2.463	0.316	0.624	3.296
$Assets_{i,t}$	331.674	585.729	11.478	56.939	1155.525
$Commercial\ NPW(Mil.)_{i,t}$	18.830	27.810	0.784	6.223	57.186
$Commercial\ Stringent_{i,t}$	10.892	10.858	0.000	7.452	27.026
$Personal\ NPW(Mil.)_{i,t}$	137.814	257.964	0.000	12.143	487.127
$Personal\ Stringent_{i,t}$	27.744	21.189	0.000	29.718	57.100
Observations	3,899				

Panel B: Univariate Differences of the Treatment and Control Groups

	Control		Treat		Diff
	Mean	SD	Mean	SD	*diff.*
General Exp Ratio	0.16	0.09	0.17	0.10	0.01***
Stringent Form	1.00	0.00	0.39	0.49	−0.61***
I(Stringent Rate)	0.35	0.48	0.08	0.27	−0.27***
Net Premiums Written	5.52	10.03	6.63	18.97	1.10
$Loss\ Volatility_{l,t}$	1.56	2.44	1.58	2.57	0.02
$Assets_{i,t}$	360.28	596.03	207.06	520.90	−153.22***
$Commercial\ NPW(Mil.)_{i,t}$	20.12	28.37	13.19	24.44	−6.93***
$Commercial\ Stringent_{i,t}$	12.25	10.46	4.98	10.61	−7.26***
$Personal\ NPW(Mil.)_{i,t}$	150.97	259.14	80.51	244.83	−70.46***
$Personal\ Stringent_{i,t}$	28.71	21.08	23.55	21.17	−5.16***
Observations	3,171		728		3,899

Notes: Panel A reports summary statistics of the single-state commercial (conventional) firm-line-year observations used in the estimation of Table 5 Column (1). Panel B reports the mean and standard deviations of the single-state commercial (conventional) firm-line-year observations used in the estimation of Table 5 Column (1) by treatment. The treatment group (Treat) are the firm-lines that operate in states that de-regulate, while the control group (Control) are the firm-lines that do not de-regulate (always stringent). The table also reports the univariate mean differences of the treatment and control groups (Diff). The tests of mean differences assume unequal variance.

Table A10: Descriptive Statistics of the Stacked Panels - Personal Lines

Panel A: Summary Statistics					
	Mean	SD	10^{th}	50^{th}	90^{th}
General Exp Ratio	0.185	0.140	0.073	0.156	0.312
Stringent Form	0.956	0.205	1.000	1.000	1.000
I(Stringent Rate)	0.382	0.486	0.000	0.000	1.000
Net Premiums Written	8.100	35.824	0.613	2.617	13.820
$Loss\ Volatility_{l,t}$	0.312	0.197	0.161	0.281	0.433
$Assets_{i,t}$	77.222	198.726	5.004	19.523	155.717
$Commercial\ NPW(Mil.)_{i,t}$	6.337	20.101	0.000	1.141	13.091
$Commercial\ Stringent_{i,t}$	6.079	9.015	0.000	0.000	19.694
$Personal\ NPW(Mil.)_{i,t}$	18.161	56.329	0.910	6.094	38.492
$Personal\ Stringent_{i,t}$	30.521	22.967	4.492	25.583	61.570
Observations	1,820				

Panel B: Univariate Differences of the Treatment and Control Groups					
	Control		Treat		Diff
	Mean	SD	Mean	SD	*diff.*
General Exp Ratio	0.19	0.14	0.18	0.10	−0.01
Stringent Form	1.00	0.00	0.43	0.50	−0.57***
I(Stringent Rate)	0.41	0.49	0.00	0.00	−0.41***
Net Premiums Written	7.93	37.16	10.18	10.50	2.26*
$Loss\ Volatility_{l,t}$	0.31	0.20	0.29	0.13	−0.02*
$Assets_{i,t}$	76.59	204.99	84.82	95.77	8.24
$Commercial\ NPW(Mil.)_{i,t}$	6.41	20.72	5.51	10.03	−0.90
$Commercial\ Stringent_{i,t}$	6.44	9.25	1.80	3.22	−4.63***
$Personal\ NPW(Mil.)_{i,t}$	17.24	58.01	29.26	27.25	12.03***
$Personal\ Stringent_{i,t}$	31.61	22.69	17.50	22.36	−14.11***
Observations	1,680		140		1,820

Notes: Panel A reports summary statistics of the single-state personal firm-line-year observations used in the estimation of Table 6 Column (1). Panel B reports the mean and standard deviations of the single-state personal firm-line-year observations used in the estimation of Table 6 Column (1) by treatment. The treatment group (Treat) are the firm-lines that operate in states that de-regulate, while the control group (Control) are the firm-lines that do not de-regulate (always stringent). The table also reports the univariate mean differences of the treatment and control groups (Diff). The tests of mean differences assume unequal variance.

Table A11: The Effect of De-regulating Forms in Commercial Lines, Sensitivity Test

Panel A: Seven-year Window, -3 to +3 years, excluding 1995			
	Full	Matched	
	(1)	(2)	(3)
Deregulate X Post	-0.0128**	-0.0165***	-0.0165***
	(0.0059)	(0.0032)	(0.0032)
Dep.Var. Mean	0.158	0.154	0.154
Event Year X Firm X Line FE	Yes	Yes	Yes
Event Year X Firm FE	No	No	Yes
Event Year X Year FE	Yes	Yes	Yes
Within R^2	0.002	0.013	0.009
N	3,528	1,372	1,372

Panel B: Eight-year Window, -3 to +4 years (same events as Seven-year Window)			
	Full	Matched	
	(1)	(2)	(3)
Deregulate X Post	-0.0151**	-0.0170***	-0.0170***
	(0.0055)	(0.0040)	(0.0040)
Dep.Var. Mean	0.156	0.151	0.151
Event Year X Firm X Line FE	Yes	Yes	No
Event Year X Firm FE	No	No	Yes
Event Year X Year FE	Yes	Yes	Yes
Within R^2	0.003	0.014	0.010
N	4,136	1,536	1,536

Notes: Robust standard errors, reported in parentheses, are clustered at the state. *$p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A12: Summary Statistics of the Product and Rate Filings Data

Panel A: Summary Statistics					
	Mean	SD	10^{th}	50^{th}	90^{th}
Average Days till Disposition$_{i,l,s,t}$	25.962	35.518	1.000	14.286	61.000
Total Form Submissions$_{i,l,s,t}$	3.677	4.080	1.000	2.000	8.000
Firm-line-state Size$_{i,l,s,t}$ *(Mil.)*	7.316	43.938	0.046	0.811	12.012
Firm-line Size$_{i,l,t}$ *(Mil.)*	10.888	66.295	0.042	1.161	18.102
Firm Size$_{i,t}$ *(Mil.)*	4659.215	10249.060	123.614	1016.993	12150.666
I(Stringent Rate$_{l,s,t}$*)*	0.141	0.348	0.000	0.000	1.000
All filings$_{l,s,t}$	1083.826	862.986	289.000	813.000	2362.000
State GDP$_{s,t}$ *(Mil.)*	0.342	0.384	0.054	0.239	0.735
State Population$_{s,t}$ *(Mil.)*	6.858	7.019	0.984	4.813	12.960
Observations	253,057				

Panel B: Univariate Differences by Stringent Form					
	Not Stringent		Stringent		Diff
	Mean	SD	Mean	SD	*diff.*
Average Days till Disposition$_{i,l,s,t}$	23.63	31.63	28.86	39.63	5.23***
Total Form Submissions$_{i,l,s,t}$	3.90	4.24	3.40	3.85	−0.50***
Firm-line-state Size$_{i,l,s,t}$ *(Mil.)*	4.68	24.55	10.59	59.66	5.90***
Firm-line Size$_{i,l,t}$ *(Mil.)*	8.15	48.41	14.29	83.19	6.14***
Firm Size$_{i,t}$ *(Mil.)*	4700.27	10135.58	4608.25	10388.04	−92.02**
I(Stringent Rate$_{l,s,t}$*)*	0.05	0.22	0.25	0.43	0.20***
All filings$_{l,s,t}$	1156.58	810.18	993.50	916.36	−163.08***
State GDP$_{s,t}$ *(Mil.)*	0.33	0.30	0.36	0.47	0.03***
State Population$_{s,t}$ *(Mil.)*	6.55	5.44	7.24	8.57	0.69***
Observations	140,164		112,893		253,057

Notes: The table shows summary statistics of the form filings data. The data includes firm-line-state-year observations from 2007 to 2014. Panel B reports the univariate mean differences by the stringent form regulation (Diff). The tests of mean differences assume unequal variance.

Table A13: The Effect of SERFF Usage on the General Expense Ratio: First Stage Results Estimating *I(SERFF Usage)*

	Main		Firm Cluster	
	(1)	(2)	(3)	(4)
I(SERFF Mandate)	0.1969***	0.1848***	0.1969***	0.1848***
	(0.0064)	(0.0091)	(0.0089)	(0.0112)
Stringent Form X I(SERFF Mandate)		0.0249*		0.0249*
		(0.0140)		(0.0133)
Stringent Form	-0.0280***	-0.0343***	-0.0280**	-0.0343**
	(0.0107)	(0.0112)	(0.0128)	(0.0136)
Stringent Rate	-0.0087	-0.0069	-0.0087	-0.0069
	(0.0163)	(0.0163)	(0.0170)	(0.0171)
ln(No. States)	0.1015***	0.1021***	0.1015***	0.1021***
	(0.0061)	(0.0061)	(0.0060)	(0.0060)
ln(Firm-Line Size)	0.0118***	0.0117***	0.0118***	0.0117***
	(0.0033)	(0.0033)	(0.0032)	(0.0032)
Loss $Volatility_{l,t}$	0.0031***	0.0030***	0.0031***	0.0030***
	(0.0005)	(0.0005)	(0.0004)	(0.0004)
ln(Firm $Size_{i,t}$*)*	-0.0045	-0.0044	-0.0045	-0.0044
	(0.0074)	(0.0074)	(0.0084)	(0.0084)
ln(State GDP Exposure)	0.0006	0.0040	0.0006	0.0040
	(0.0414)	(0.0415)	(0.0480)	(0.0484)
ln(State Population Exposure)	-0.0034	-0.0073	-0.0034	-0.0073
	(0.0434)	(0.0435)	(0.0503)	(0.0507)
N	82,936	82,936	82,936	82,936

Notes: The table shows the first-stage results for the two-stage least squares model in equations (4)-(5). See Table 3 for definitions of other variables. Robust standard errors, reported in parentheses, are clustered at the firm-line level. *$p < 0.1$, ** $p < 0.05$, *** $p < 0.01$